

★ WEEK 10 · PART 1 of 3

Hidden Markov Models: The Market Has Moods

A Discrete Hidden State That Switches Between Regimes

Markov chains · the transition matrix · emissions · market regimes

The Kalman filter tracked a continuous hidden state. Now the hidden state is discrete — a regime like calm or crisis — that the market switches between, revealed only through the returns it emits.

1

PART 1 · FROM CHAINS TO HMMs

A Hidden Regime, Emitting Returns

Markov chains and their one-step memory, the transition matrix, the leap from visible to hidden states, the four pieces of an HMM, and how regimes show up in markets.

From Pushkin to Speech Recognition — Part 1

How counting letters in a poem founded a field

In 1913, Andrey Markov analyzed the sequence of vowels and consonants in Pushkin's Eugene Onegin. He showed the letters were not independent — the next depended on the current. That was the first Markov chain.

The insight

- Successive letters carry dependence, not randomness.
- But the dependence reaches back only ONE step.
- Given the current letter, the past adds nothing.

Why it mattered

- It modeled sequences with minimal memory.
- Simple enough to compute, rich enough to be useful.
- It became the foundation of a huge field.

From Pushkin to Speech Recognition — Part 2

The hidden twist, and the lesson

Decades later, Leonard Baum and colleagues added a twist: what if the chain is HIDDEN, and you see only noisy signals of it? That hidden Markov model became the engine of speech recognition.

The twist

- The state (the spoken sound) is not directly observed.
- You see only the acoustic signal it produces.
- Infer the hidden sequence from the observations.

The lesson for us

- Markets hide their state too — calm or crisis.
- We see only the returns each regime emits.
- The same model infers the regime from the data.

From a Continuous State to a Discrete One

Where Week 9 left off

1

Kalman tracked a continuous state

Week 9 estimated a smoothly-evolving hidden state — a level, a drifting beta — with a linear-Gaussian model.

2

But some states are discrete

A market is arguably in one of a few regimes: calm or crisis, bull or bear. That state doesn't drift smoothly — it SWITCHES.

3

HMMs model the switching hidden state

A hidden Markov chain jumps between regimes, and each regime emits returns from its own distribution. Same spirit as Kalman, discrete state.

Part 1 in one line: a hidden regime switches over time and reveals itself through the returns.

Markov Chains: Memory of Just One Step

The future depends only on the present state

A Markov chain is a sequence of states where the next state depends only on the current one — not on the whole history. Once you know where you are, the past tells you nothing more.

The one-step memory

- The next state's probabilities depend only on the current state.
- The path taken to get here is irrelevant.
- This is the Markov property — memorylessness beyond one step.

Why it is powerful

- It makes long sequences tractable to model and compute.
- A few transition probabilities describe the whole process.
- It is simple, yet captures real persistence.

The Markov Property

One line that defines the whole idea

$$\text{Markov property: } P(X_{t+1} \mid X_t, X_{t-1}, \dots, X_1) = P(X_{t+1} \mid X_t)$$

Reading it

- The probability of the next state, given all history...
- ...equals its probability given only the current state.
- The conditioning on the past collapses to the present.

The intuition

- The present state is a sufficient summary of the past.
- It carries everything relevant for the future.
- That is exactly what makes the chain memoryless.

The Transition Matrix

All the switching probabilities in one grid

$$A_{ij} = P(X_{t+1} = j \mid X_t = i), \quad \sum_j A_{ij} = 1 \quad (\text{each row sums to one})$$

What it is

- Entry (i, j) is the probability of moving from state i to state j.
- Each row is a probability distribution — it sums to one.
- It fully describes the chain's dynamics.

Reading the diagonal

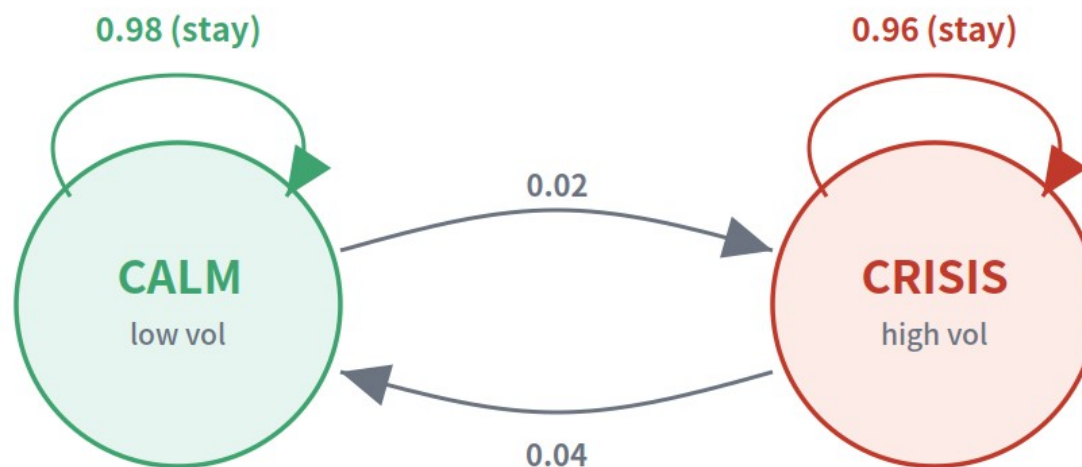
- Large diagonal entries mean the state tends to STAY.
- That persistence is called stickiness.
- Sticky regimes produce long calm and crisis spans.

A Two-State Markov Chain

The transition matrix, drawn. High stay-probabilities (the self-loops) make each regime persistent; the small cross-arrows are the occasional switches between calm and crisis.

A Two-State Markov Chain

The transition matrix, drawn: probabilities of staying vs switching.



Regimes are STICKY

High stay-probabilities mean each regime persists - so returns cluster into long calm and crisis spans.

Sticky Regimes and the Long Run

Persistence, and where the chain settles

Stickiness

- High stay-probabilities keep the chain in a regime.
- So states persist rather than flickering each step.
- This is why volatility clusters into regimes.

The stationary distribution

- Over the long run, the chain spends a fixed fraction of time in each state.
- That fraction is the stationary distribution.
- It tells you how common calm vs crisis is overall.

From Visible to Hidden

We don't get to see the state

In a plain Markov chain you observe the state directly. The crucial leap of an HMM: the state is HIDDEN. You see only observations it emits, and must infer the state behind them.

Plain Markov chain

- The state itself is observed at each step.
- You can read the transitions directly.
- Nothing is hidden.

Hidden Markov model

- The state is latent — never observed.
- Each state emits an observation you DO see.
- The whole challenge is inferring the hidden state.

The Pieces of an HMM

What fully defines the model

an HMM = (π, A, B) : initial π , transition A , emissions B

The three parts

- An initial distribution over the starting state.
- A transition matrix for switching between states.
- An emission model — each state's observation distribution.

Plus one choice

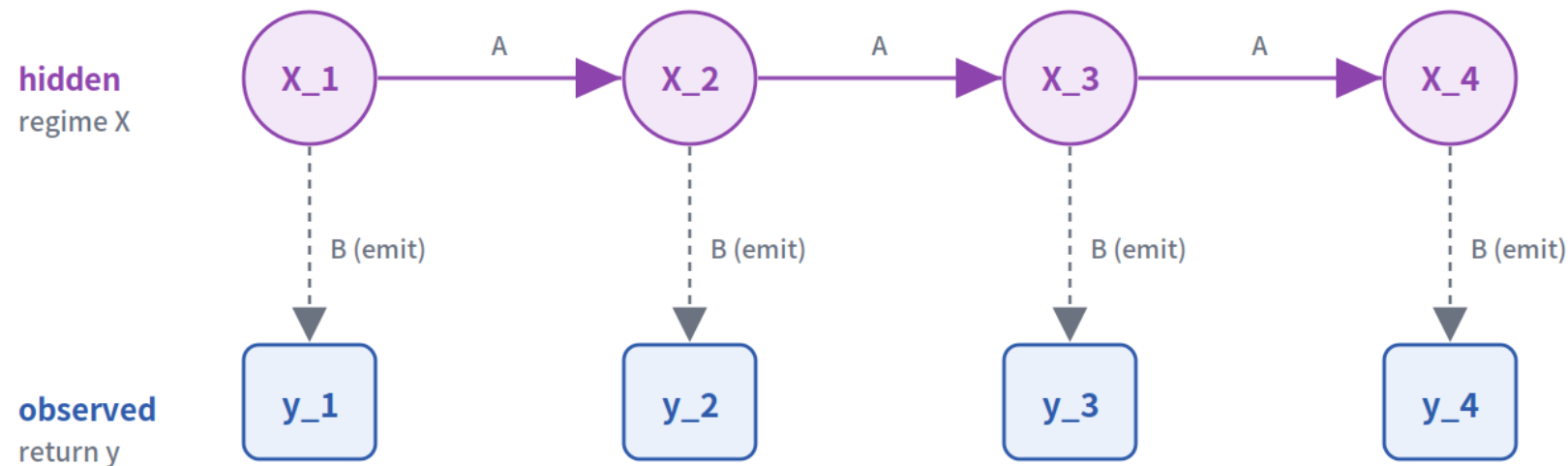
- The NUMBER of hidden states (e.g. two: calm, crisis).
- That count is yours to choose (Part 3).
- Together these fully specify the HMM.

The HMM Picture

A hidden regime chain switches by the transition A; each regime emits an observed return by the emission model B. We see only the bottom row and infer the hidden regimes above.

The HMM Picture

A hidden regime chain switches by A and emits the returns we observe by B.



The regime is never observed. Its distribution (calm vs crisis) shapes the returns - and we infer it from them.

Emissions: Each Regime Its Own Distribution

How a hidden state shows up in the data

$$\text{emission (Gaussian): } y_t \mid (X_t = i) \sim \mathcal{N}(\mu_i, \sigma_i^2)$$

The idea

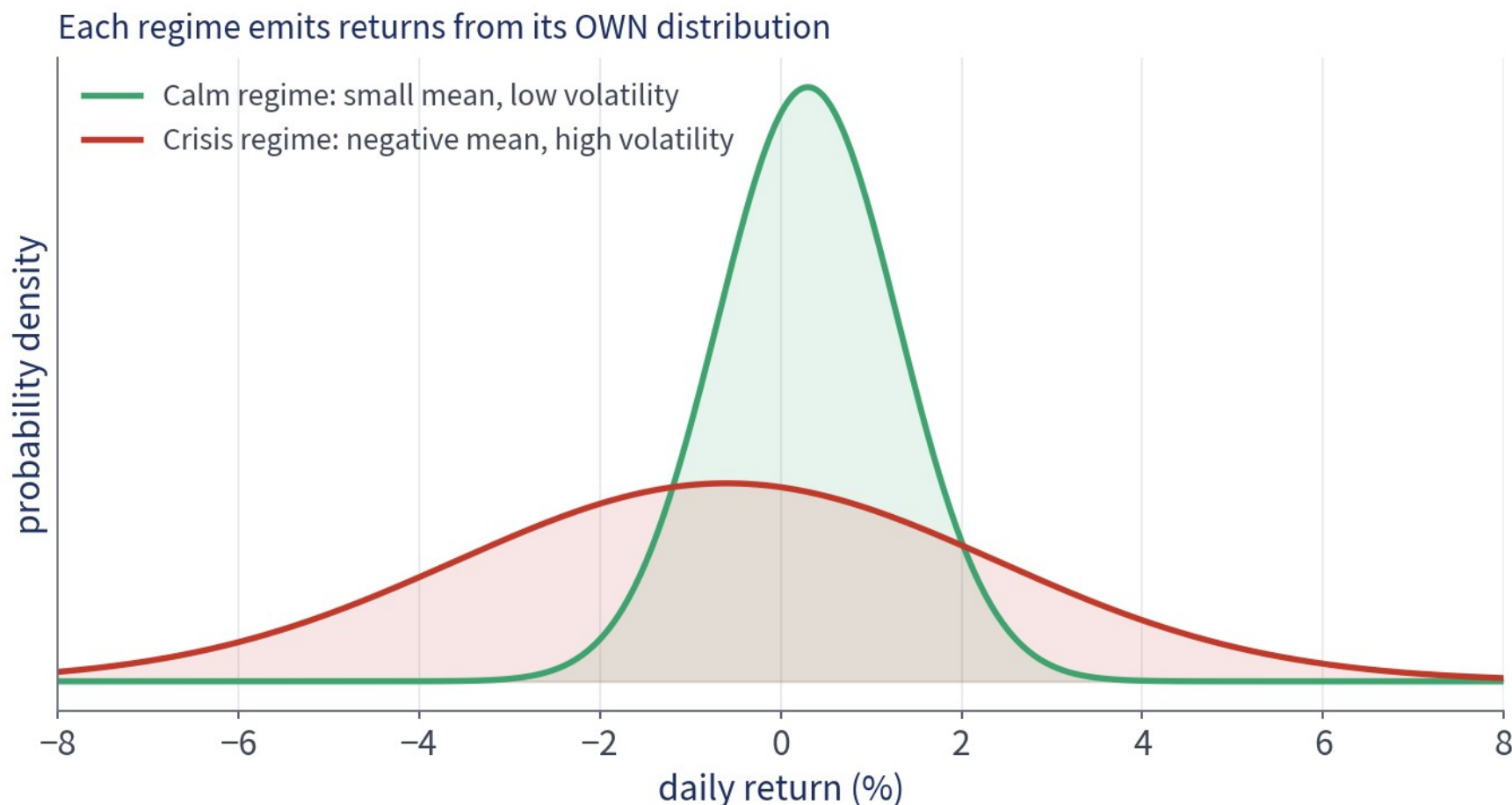
- Each regime emits observations from its own distribution.
- A calm regime: small mean, low variance returns.
- A crisis regime: negative mean, high variance returns.

Why this reveals the state

- Different regimes leave different fingerprints on the data.
- A run of large, wild returns hints at a crisis regime.
- The emission model is the bridge from state to observation.

Two Regimes, Two Distributions

The calm regime emits a tight distribution of small returns; the crisis regime emits a wide one with a negative mean. Which regime you're in shapes what returns you tend to see.



Regimes in Markets

The canonical financial use of HMMs

The natural financial reading: the market is in a hidden regime — calm or crisis, bull or bear — and returns are the emissions. The HMM infers which regime is active from the returns alone.

The regimes

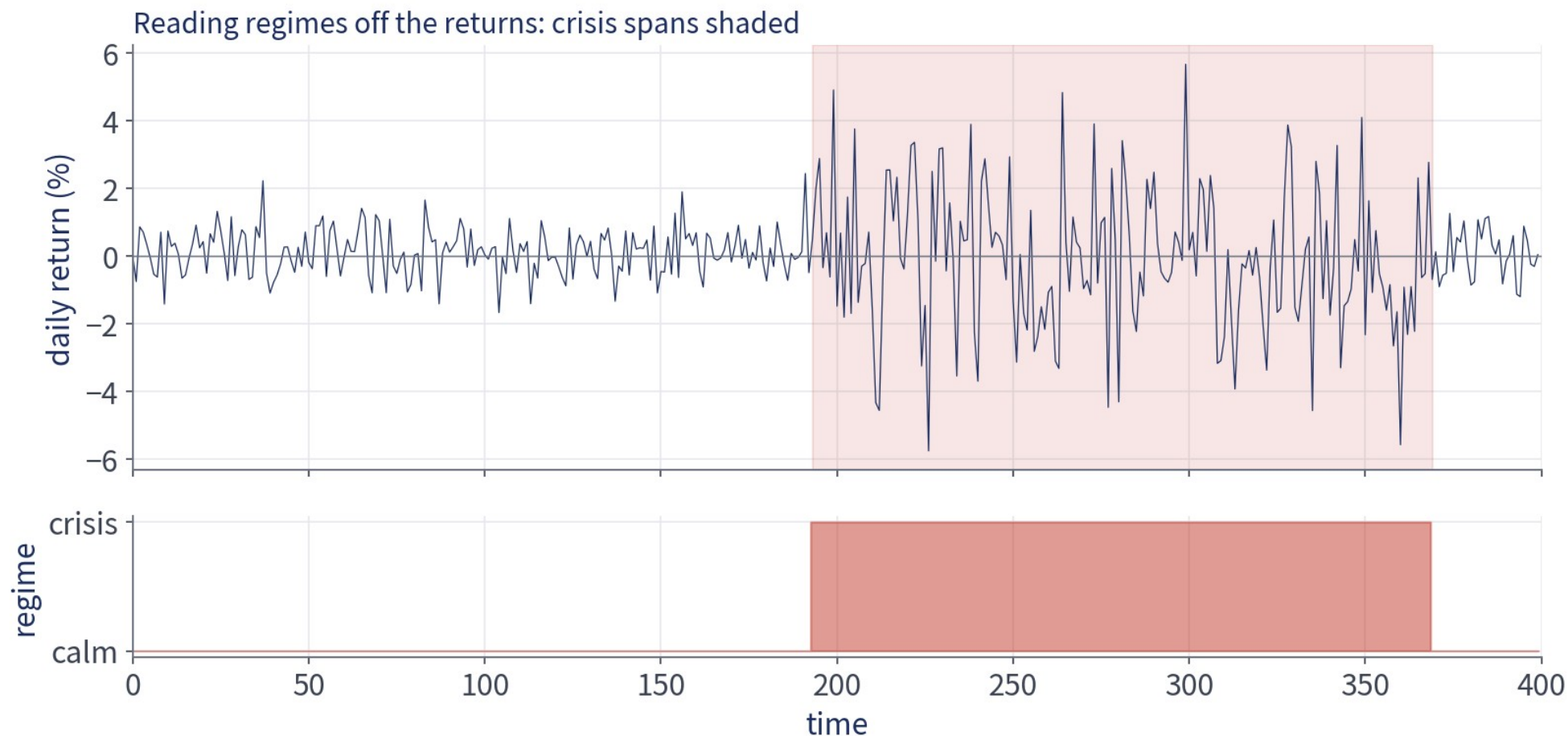
- Calm: modest positive drift, low volatility.
- Crisis: negative drift, high volatility, fat tails.
- Each is a state of the hidden chain.

What the HMM gives

- A probability of being in each regime, each day.
- A decoded regime path over history.
- An early read when the market shifts mood.

Reading Regimes Off the Returns

A return series with crisis spans shaded and the decoded regime below. Calm periods are quiet; crisis periods are volatile and clustered — exactly what a sticky, two-state HMM captures.



Why HMMs Fit Markets

Regimes match what returns actually do

What returns show

- Volatility clusters — quiet stretches and stormy stretches.
- Fat tails — occasional extreme moves.
- Persistence — turbulence tends to beget turbulence.

How HMMs capture it

- Sticky regimes produce the clustering directly.
- A high-volatility regime supplies the fat tails.
- A discrete switch matches sudden shifts in mood.

HMM vs Kalman Filter

Discrete state vs continuous state

Aspect	Kalman filter (Week 9)	HMM (Week 10)
Hidden state	continuous	discrete — a few regimes
State dynamics	linear-Gaussian transition	a transition matrix
Emissions	linear plus Gaussian noise	one distribution per regime
Inference	predict-update recursion	forward / Viterbi / Baum-Welch

Same core idea — a hidden state seen through noisy data — but the state is now discrete.

The Generative Story

How an HMM produces a return series

The recipe

- Pick a starting regime from the initial distribution.
- Emit a return from that regime's distribution.
- Switch regime by the transition matrix, then repeat.

Why it helps

- It tells you exactly what the model assumes.
- Inference simply reverses this story.
- You can simulate synthetic regime data to test methods.

What HMMs Assume

Two assumptions to keep in mind

$P(y_t \mid X_t)$ depends only on the current regime X_t (conditional independence)

The two assumptions

- Markov: the regime's next step depends only on the current regime.
- Conditional independence: each return depends only on its own regime.
- Given the regime, the observation carries no extra memory.

Why they matter

- They are what make the algorithms efficient (Part 2).
- They are approximations — real markets bend them.
- Knowing them tells you when an HMM may mislead.

What Parts 2 and 3 Add

From model to method to use

Part 2 — the algorithms

- Evaluation: the forward algorithm for the likelihood.
- Decoding: Viterbi for the most likely regime path.
- Learning: Baum-Welch to fit a model with hidden states.

Part 3 — in practice

- How many regimes to choose, and the pitfalls.
- Pairs trading Stage IV: trade only in favorable regimes.
- Cautions, and the bridge to continuous-time models.

Fit a 2-State Gaussian HMM

Direct Claude Code to detect calm and crisis

GOAL Fit a two-state Gaussian HMM to a return series and see the calm and crisis regimes it recovers.

[INPUT] Prompt · Claude Code

Write and run `hmm_regimes.py` that:

1. loads a daily return series (an index is fine);
2. fits a 2-state Gaussian HMM (e.g. `hmmlearn GaussianHMM` with 2 components);
3. prints each regime's mean and volatility, and the transition matrix (note the stickiness);
4. plots the returns with the decoded regimes shaded (calm vs crisis).

Tell me which state is 'crisis' (higher vol, negative mean) and how persistent each regime is.

How to work with it

- The crisis state has higher volatility.
- Diagonal transition entries should be near 1.
- Regimes should form long clustered spans.
- Keep it; Part 2 explains how it's fit.

Deliverable: `hmm_regimes.py` + regime parameters and a shaded return plot.

Three Questions Before You Move On

If you can answer these, Part 1 has landed

Q1 In a Markov chain, what does the next state depend on?

Only the current state

Q2 What makes a Markov model 'hidden'?

The state is unobserved

Q3 How does a hidden regime reveal itself in the data?

Through its emissions

All three come from the Markov property, the hidden leap, and the emission model.

Three Misconceptions to Avoid

Where HMM intuition slips

Common myths

- "The regimes are labeled in the data."
- "An HMM needs long memory to model persistence."
- "The number of regimes is given by the data."

The reality

- Regimes are latent — the HMM infers them, unlabeled.
- One-step memory plus sticky transitions gives persistence.
- You CHOOSE the number of regimes; it's a modeling decision.

Three Things to Carry Forward

Part 1 in one screen

1

A Markov chain has one-step memory

The next state depends only on the current one, encoded in a transition matrix. Sticky (high-diagonal) chains persist, producing the clustering we see in markets.

2

An HMM hides that chain

The state is latent; we see only observations each regime emits. An HMM is an initial distribution, a transition matrix, and an emission model per state.

3

In markets, the state is the regime

Calm and crisis emit different return distributions. The HMM is the discrete-state cousin of the Kalman filter — same idea, a switching hidden state.

Next — Part 2: the three problems and their algorithms — forward, Viterbi, and Baum-Welch.

★ WEEK 10 · PART 1 · PRACTICE

Markov Regimes, Worked by Hand

Persistence and the Long Run; Which Regime Made This Return

The transition matrix and the emissions, made concrete

Part 1's foundations, with the arithmetic shown. We read a regime transition matrix for duration and long-run mix, then use emission distributions and one Bayes step to infer which hidden regime a return most likely came from.

1P

PART 1 · APPLICATIONS

Two Regime Calculations, by Hand

Case A: a calm/turbulent transition matrix, worked for expected duration, a two-step probability, and the long-run mix. Case B: emission distributions and one Bayes step to infer the hidden regime from a return. Then a Claude Code lab.

What Part 1 Told Us

Now we make it concrete - with numbers

1

A Markov chain has one-step memory

The next regime depends only on the current one. All the switching probabilities live in a transition matrix.

2

Regimes are sticky, and settle

High stay-probabilities make regimes persist; over the long run the chain spends a fixed fraction of time in each.

3

The state is hidden; emissions reveal it

Each regime has its own return distribution. A return is a clue - and one Bayes step turns it into a regime probability.

Everything here is a calculator exercise: one transition matrix, two return distributions, one bad day.

How These Cases Work

Data, then arithmetic, then meaning

Each case has three steps

- Start from the matrix or the distributions.
- Compute duration, mix, or a posterior by hand.
- Then read what it says about the market.

What you should watch

- How stickiness sets the expected duration.
- How the chain settles to a long-run mix.
- How one return shifts the regime belief.

Reading the Transition Matrix

Persistence, and where the chain settles

Two regimes: a calm market and a turbulent one. The transition matrix holds the switching probabilities. From it alone - no data yet - we can read how long each regime lasts, the chance of switching over two steps, and the fraction of time the market spends turbulent in the long run

The matrix

- Calm stays calm 95% of days.
- Turbulent stays turbulent 80%.
- Calm is stickier than turbulence.

What we'll do by hand

- Expected duration of each regime.
- The two-step transition probability.
- The long-run stationary mix.

The Transition Matrix

All the switching in one grid

$$P = \begin{pmatrix} 0.95 & 0.05 \\ 0.20 & 0.80 \end{pmatrix} \quad \begin{array}{l} \text{row} = \text{from Calm / Turbulent} \\ \text{col} = \text{to Calm / Turbulent} \end{array}$$

How to read it

- Each row is where you are now.
- Each column is where you go next.
- Rows sum to one - you go somewhere.

What it encodes

- The diagonal is the stay-probability.
- The off-diagonal is the switch chance.
- That's the whole dynamics of the regimes.

Expected Duration

How long a regime lasts

$$\text{expected duration} = \frac{1}{1-p_{\text{stay}}} : \quad \text{Calm} = \frac{1}{1-0.95} = \mathbf{20} \text{ days}, \quad \text{Turbulent} = \frac{1}{1-0.80} = \mathbf{5} \text{ days}$$

The formula

- Expected duration = $1 / (1 - \text{stay probability})$.
- It's the mean of a geometric distribution.
- Higher stickiness, longer spells.

The numbers

- Calm lasts about 20 days on average.
- Turbulence lasts about 5 days.
- Calm periods are four times longer.

A Two-Step Transition

Probabilities compound

$$P(\text{Calm} \rightarrow \text{Calm in 2}) = 0.95(0.95) + 0.05(0.20) = \mathbf{0.9125}$$

$$P(\text{Calm} \rightarrow \text{Turb in 2}) = 0.95(0.05) + 0.05(0.80) = 0.0875$$

To be calm in two days you can stay calm both days, or dip into turbulence and come back. Summing the paths gives 0.9125 - the entries of the squared matrix.

The Long-Run Mix

Where the chain settles

$$\pi_C (0.05) = \pi_T (0.20), \quad \pi_C + \pi_T = 1 \Rightarrow \pi_C = \mathbf{0.80}, \quad \pi_T = \mathbf{0.20} \text{ (long-run mix)}$$

The balance condition

- In steady state, flow out of calm equals flow in.
- So π_C times 0.05 equals π_T times 0.20.
- With the two summing to one.

The answer

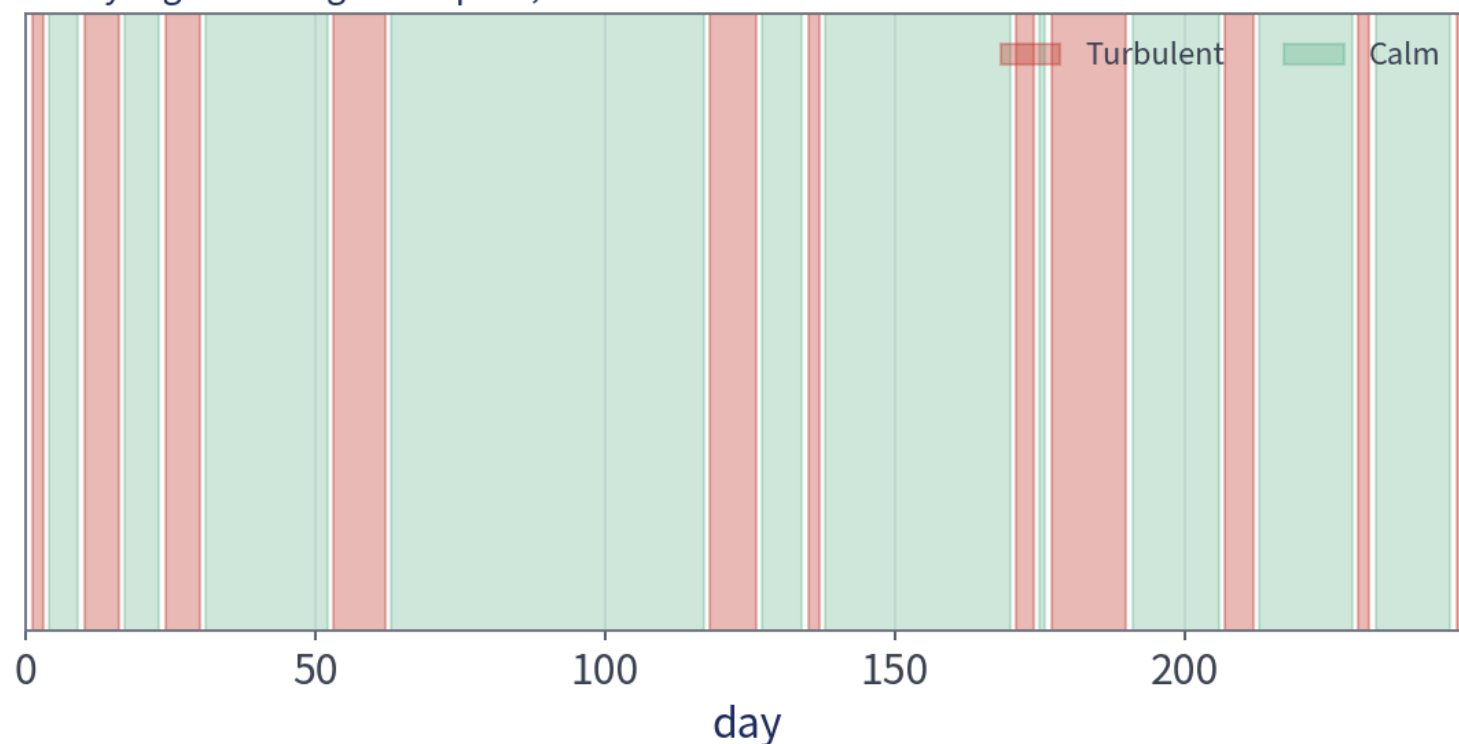
- The market is calm 80% of the time,
- and turbulent 20%.
- Independent of where it starts.

Sticky Regimes, a Steady Mix

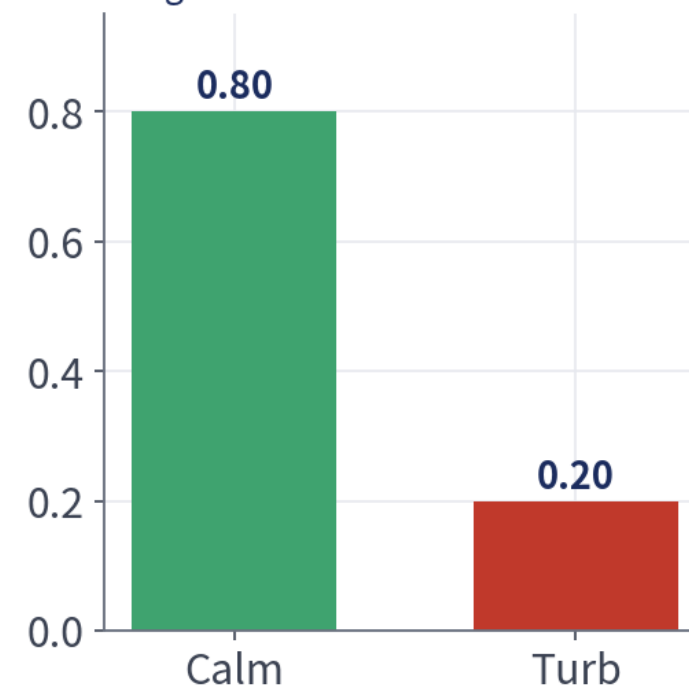
A simulated path shows the signature of sticky regimes: long green calm stretches broken by short red turbulent bursts. Over the long run, the fractions settle exactly to the stationary mix - 80% calm, 20% turbulent - no matter where the chain began

Case A: 80% calm / 20% turbulent in the long run - calm lasts 20 days, turbulence 5

Sticky regimes: long calm spells, short turbulent bursts



Long-run mix



What the Matrix Tells You

Structure, before any data

You already know

- How long a calm or turbulent spell tends to last.
- The odds of switching over any horizon.
- The long-run share of turbulent days.

Why it matters

- It sets expectations for regime persistence.
- It calibrates how fast to react to a switch.
- It's the backbone the emissions hang on.

Which Regime Made This Return?

Emissions, and one Bayes step

The regime is hidden - but each one emits returns with its own distribution. Calm returns are small and positive; turbulent returns are negative and wild. Given a -2% day, we compute how likely that return is under each regime, then combine with the prior to get the probability we're now in the turbulent regime

The emissions

- Calm: returns $\sim N(+0.5\%, 1\%)$.
- Turbulent: returns $\sim N(-1\%, 3\%)$.
- Today's return is -2.0%.

What we'll do by hand

- Compute the likelihood under each regime.
- Multiply by the prior regime probabilities.
- Normalize to a posterior.

The Likelihood of the Return

How well each regime explains -2%

$$\text{Calm} \sim N(+0.5, 1.0^2) : \ell_C = \varphi\left(\frac{-2.0-0.5}{1.0}\right) = \varphi(-2.5) = 0.018$$

$$\text{Turb} \sim N(-1.0, 3.0^2) : \ell_T = \varphi\left(\frac{-2.0+1.0}{3.0}\right) = \varphi(-0.33) = 0.126$$

A -2% return is 2.5 standard deviations below the calm mean but well within the turbulent spread - so its density is about seven times higher under turbulence.

One Bayes Step

From likelihood to regime probability

$$P(\text{Turb} \mid r) = \frac{\pi_T \ell_T}{\pi_C \ell_C + \pi_T \ell_T} = \frac{0.20(0.126)}{0.80(0.018) + 0.20(0.126)} = \mathbf{0.64}$$

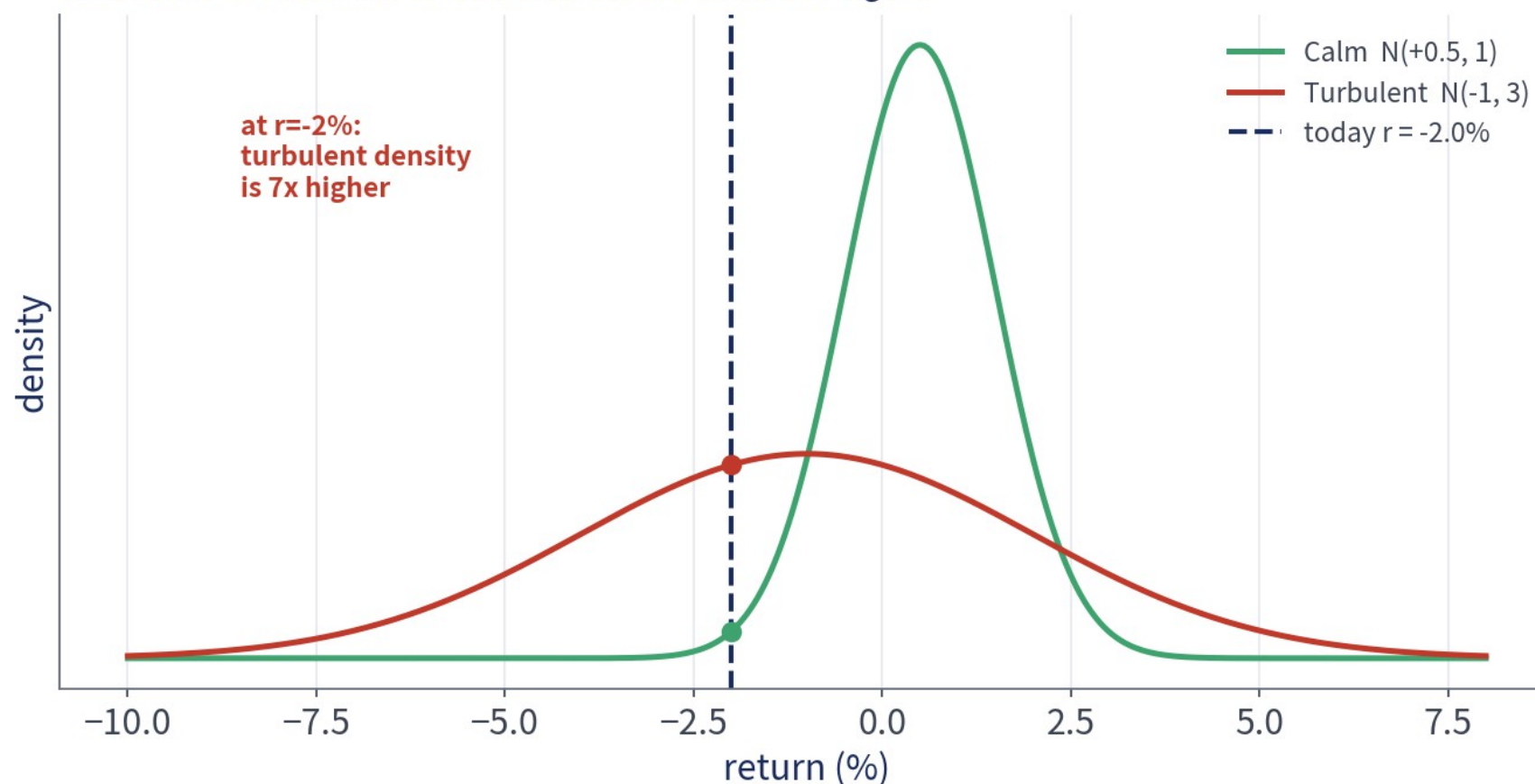
$\Rightarrow$ a -2% day lifts turbulent belief from 0.20 to 0.64

Combining the prior (20% turbulent) with the likelihoods lifts the posterior to 0.64 - the single bad day more than triples our belief that we're in turbulence.

The Return Points to a Regime

At -2%, the calm distribution (green) has almost no density - it's a 2.5-sigma event there - while the turbulent one (red) is comfortably wide. That mismatch in density is what Bayes turns into a posterior, shifting belief sharply toward turbulence

Case B: the -2% return is far likelier under the turbulent regime



Reading the Hidden State

A return is evidence, not proof

What one return gives

- Not certainty - a shifted probability.
- A big loss is stronger evidence than a small one.
- The prior still matters, especially early.

Why this is the seed

- This is the emission half of the HMM.
- Chaining it over time is the forward algorithm.
- That's exactly where Part 2 goes.

The Two Ingredients of an HMM

Transitions and emissions

An HMM is just these two pieces. The transition matrix says how regimes evolve - their stickiness and long-run mix. The emissions say how each regime shows up in returns. Case A worked the transitions; Case B worked the emissions and one inference step.

Together they define the whole model

Transitions (Case A)

- How the hidden regime moves.
- Duration and the long-run mix.
- The backbone of persistence.

Emissions (Case B)

- How each regime generates returns.
- The likelihood of an observation.
- The clue that reveals the hidden state.

Reproduce the Regime Calcs

Direct Claude Code to check your arithmetic

GOAL Verify the transition-matrix reads and the Bayes step, then simulate a regime path.

[INPUT] Prompt · Claude Code

Write and run part1_practice.py that:

CASE A - transitions:

- * from $P = \begin{bmatrix} 0.95 & 0.05 \\ 0.20 & 0.80 \end{bmatrix}$, print each regime's expected duration = $1/(1-\text{stay})$;
- * print P squared and the stationary distribution;
- * simulate a path and check the long-run mix.

CASE B - emissions:

- * with $\text{Calm} \sim N(0.5, 1)$ and $\text{Turb} \sim N(-1, 3)$, print the density of $r = -2$ under each;
- * apply Bayes with prior $0.80/0.20 \rightarrow$ posterior.

Match every printed number to the slides.

Check against the slides

- duration 20 and 5 days.
- two-step $C \rightarrow C = 0.9125$.
- stationary $0.80 / 0.20$.
- posterior turbulent = 0.64.

Deliverable: part1_practice.py whose printed numbers match the transition and emission calcs.

Three Questions on the Cases

If you can answer these, the practice has landed

Q1 How do you get a regime's expected duration?

$1 / (1 - \text{stay probability})$

Q2 What is the stationary distribution?

The long-run fraction of time in each regime

Q3 How does a return update the regime belief?

Bayes: prior times emission likelihood

All three come from the transition matrix and the emission-plus-Bayes step.

Three Pitfalls to Avoid

Where regime reasoning slips

Common mistakes

- "A high stay-probability means it never switches."
- "One bad return proves we're in a crisis."
- "The starting regime sets the long-run mix."

The reality

- It switches - just rarely; duration is $1/(1-\text{stay})$.
- It's evidence, not proof - Bayes gives a probability.
- The stationary mix is independent of the start.

Two Calculations to Remember

Part 1 applications in one screen

1

Read the transition matrix

Stay-probabilities gave expected durations of 20 days (calm) and 5 days (turbulent); the two-step calm-to-calm probability was 0.9125; and the long-run mix settled at 80% calm, 20% turbulent - all from the matrix alone.

2

Infer the hidden regime from a return

A -2% day had density 0.018 under calm but 0.126 under turbulence. One Bayes step, starting from a 20% prior, lifted the posterior probability of turbulence to 0.64 - the return shifted belief, but didn't prove it.

3

Transitions plus emissions define the HMM

Case A is how the regime moves; Case B is how it shows up in returns. Chaining the emission step over time is the forward algorithm - exactly where Part 2 begins.

Next — Part 2 practice: the forward algorithm, Viterbi, and Baum-Welch, run by hand.

★ WEEK 10 · PART 2 of 3

The Three Problems and Their Algorithms

Evaluation, Decoding, and Learning

The forward algorithm · Viterbi · Baum-Welch · filtering vs smoothing

An HMM is only useful once you can compute with it. Three questions — how likely, which path, which parameters — are answered by three classic dynamic-programming algorithms over a trellis of states.

2

PART 2 · ALGORITHMS

Computing With Hidden States

Evaluation by the forward algorithm, decoding by Viterbi, learning by Baum-Welch, and the filtering-versus-smoothing distinction — now applied to discrete regimes.

We Have the Model — Now, the Computation

Part 1 built the HMM; Part 2 makes it work

1

The model is defined

Part 1 gave the pieces: an initial distribution, a transition matrix, and an emission per regime. But we never said how to actually compute anything.

2

Three questions arise

How likely is the data? Which regime path is most likely? And which parameters fit best? Everything you do with an HMM is one of these three.

3

Three algorithms answer them

The forward algorithm, Viterbi, and Baum-Welch — all efficient dynamic programs over a trellis of states across time.

Part 2 in one line: evaluate, decode, and learn — three problems, three trellis algorithms.

The Three Problems of an HMM

Everything you do is one of these

Every HMM task falls into one of three problems: evaluation (how likely is the data?), decoding (what was the hidden path?), and learning (what parameters fit?). Each has its own algorithm.

The three questions

- Evaluation: the likelihood of the observed data.
- Decoding: the most likely sequence of hidden regimes.
- Learning: the parameters that best explain the data.

Why frame it this way

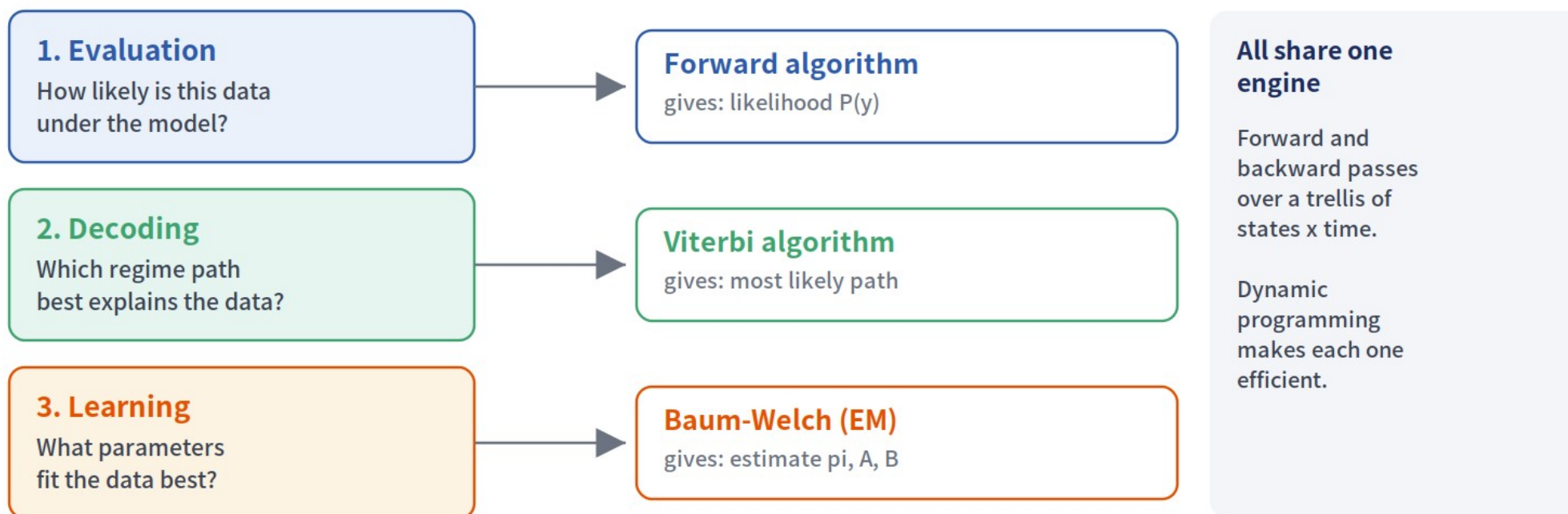
- It organizes everything HMMs can do.
- Each maps to one efficient algorithm.
- They build on each other — learning uses evaluation inside.

Three Questions, Three Algorithms

Evaluation is answered by the forward algorithm, decoding by Viterbi, and learning by Baum-Welch. All three are dynamic programs over the same trellis of states across time.

Three Questions, Three Algorithms

Everything you do with an HMM is one of these three problems.



Evaluation: How Likely Is This Data?

The likelihood under the model

The question

- Given a fitted HMM, how probable is an observed return series?
- This is the model's likelihood for that data.
- It is the basis for comparing and fitting models.

Why it matters

- It scores how well the model explains the data.
- It drives model selection — which HMM is better.
- Learning maximizes exactly this quantity.

Why the Naive Approach Fails

Enumerating every path is hopeless

The direct way to get the likelihood is to sum over every possible hidden path. But the number of paths grows exponentially with the length of the series — utterly infeasible for real data.

The blow-up

- With K regimes and T steps, there are K -to-the- T paths.
- Even a short series has astronomically many.
- Summing them one by one is impossible.

The way out

- Don't re-enumerate — reuse shared sub-computations.
- Dynamic programming collapses the exponential sum.
- That is exactly the forward algorithm.

The Forward Algorithm

Likelihood without enumerating every path

$$\alpha_t(j) = \left[\sum_i \alpha_{t-1}(i) A_{ij} \right] b_j(y_t) \quad \alpha_t(j) = P(y_{1:t}, X_t = j)$$

The forward variable

- Alpha at time t is the joint probability of the data so far AND being in state j.
- It is built recursively from the previous column.
- Each step: combine incoming paths, then apply the emission.

Why it is efficient

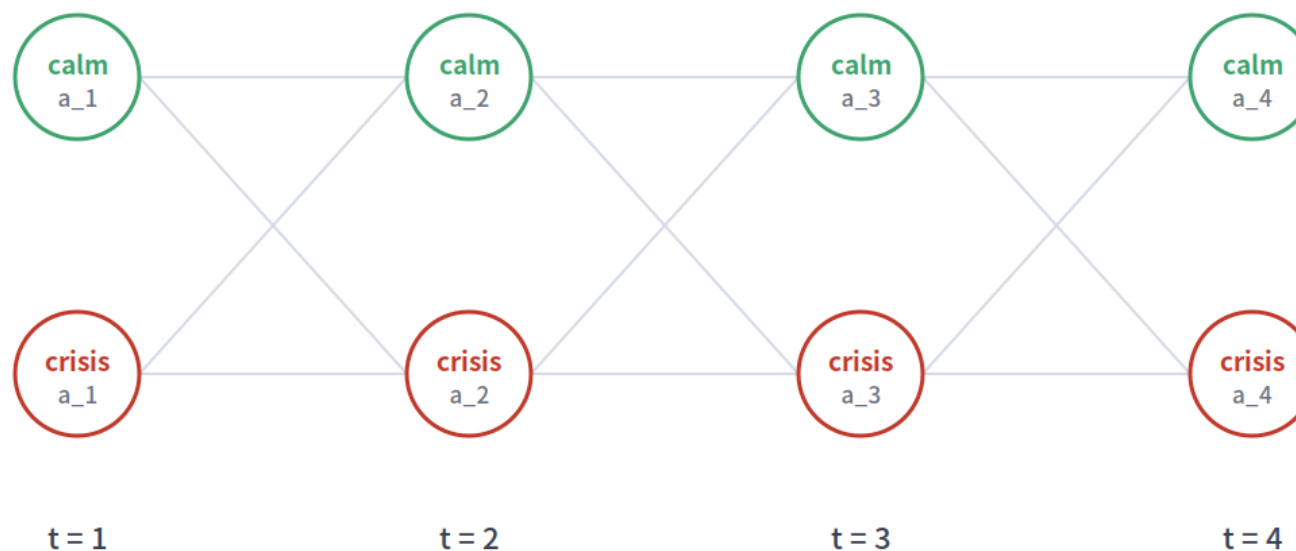
- Work grows linearly with the series length, not exponentially.
- Each column reuses the last — no re-enumeration.
- The whole exponential sum collapses to a simple recursion.

The Forward Trellis

States down, time across. Each node's value combines all incoming paths from the previous column times the transition, then the emission. The likelihood is the final column, summed.

The Forward Trellis

Sum over all paths, cheaply: each node reuses the last column instead of re-enumerating.



Each node's alpha = (sum of incoming alphas x transition) x emission. The likelihood is the last column, summed.

The Likelihood, Cheaply

Sum the final column

likelihood: $P(y_{1:T}) = \sum_i \alpha_T(i)$ (sum over paths, done cheaply)

The payoff

- Summing the forward variables at the last step gives the likelihood.
- The exponential path-sum is computed in linear time.
- One forward pass, and you have the model's score.

What it unlocks

- Comparing candidate HMMs by likelihood.
- The inner loop of Baum-Welch learning.
- The normalizer for regime probabilities.

Filtered Regime Probabilities

The probability of each regime, right now

$$\text{filtered regime prob: } P(X_t = i \mid y_{1:t}) = \frac{\alpha_t(i)}{\sum_j \alpha_t(j)}$$

What it is

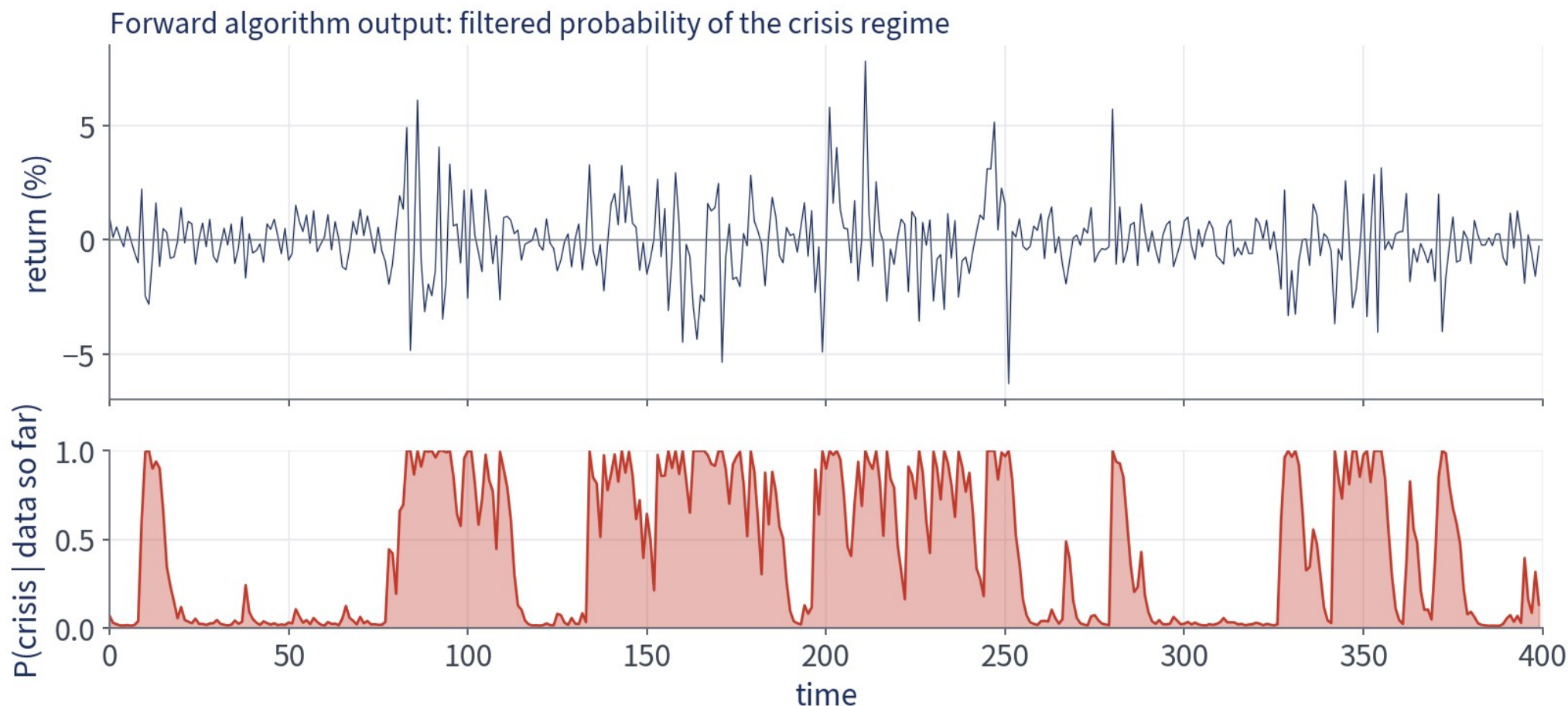
- Normalize the forward variables and you get a probability.
- It is the chance of each regime given data up to now.
- This is filtering — the Week 8 idea, for discrete states.

How you use it

- A running probability of being in the crisis regime.
- It updates causally as each return arrives.
- A live signal of the market's current mood.

Filtered Probability of Crisis

The forward algorithm gives, at each moment, the probability of being in the crisis regime given the data so far. It rises through turbulent stretches and falls back in calm ones.



Decoding: The Hidden Path

Which regime sequence explains the data?

The question

- Given the returns, what was the most likely sequence of regimes?
- Not just the probability each day — the best whole path.
- This labels history into calm and crisis stretches.

Why it matters

- It gives a single, coherent regime timeline.
- It is what you plot to 'see' the regimes.
- It underpins regime-aware decisions (Part 3).

Pointwise Best vs Single Best Path

Two different notions of 'most likely'

Pointwise (filtering/smoothing)

- Pick the most probable regime at EACH time separately.
- It can produce an inconsistent, jumpy path.
- Good for probabilities, not for a coherent sequence.

Single best path (Viterbi)

- Find the one sequence that is jointly most likely.
- It respects the transition structure throughout.
- This is the right tool for a decoded timeline.

The Viterbi Algorithm

The single most likely regime path

$$\text{Viterbi: } \delta_t(j) = \left[\max_i \delta_{t-1}(i) A_{ij} \right] b_j(y_t) \Rightarrow \text{backtrack the best path}$$

The idea

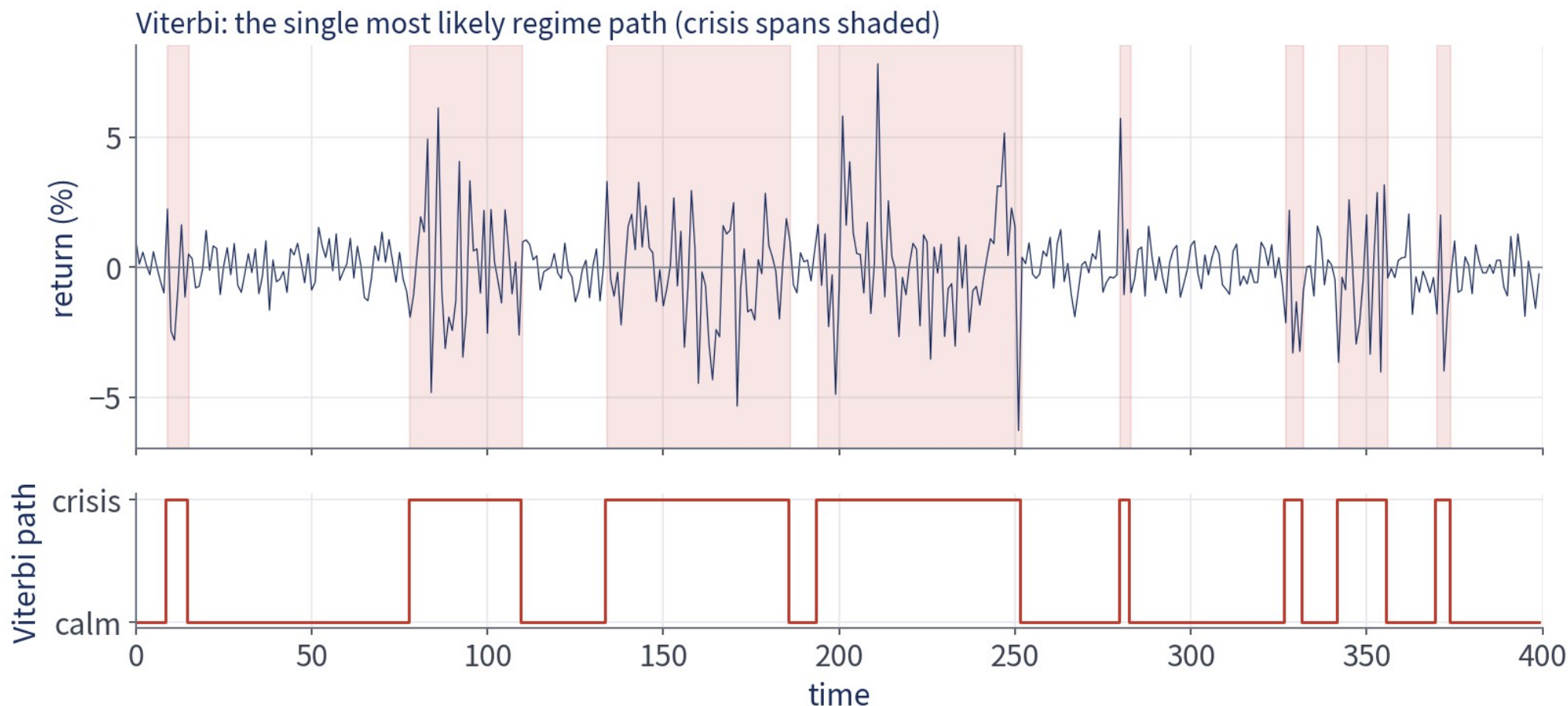
- Like the forward pass, but take a MAX instead of a sum.
- Track, at each node, the best incoming path.
- Then backtrack from the end to recover the sequence.

The result

- The globally most likely sequence of regimes.
- Coherent, transition-respecting, and unique.
- The standard way to decode an HMM.

The Viterbi Path

The single most likely regime sequence, decoded from the returns. Crisis spans (shaded) form coherent blocks — Viterbi respects the sticky transitions rather than flickering day to day.



Filtering vs Smoothing, for Regimes

The Week 8 distinction, now discrete

Filtering

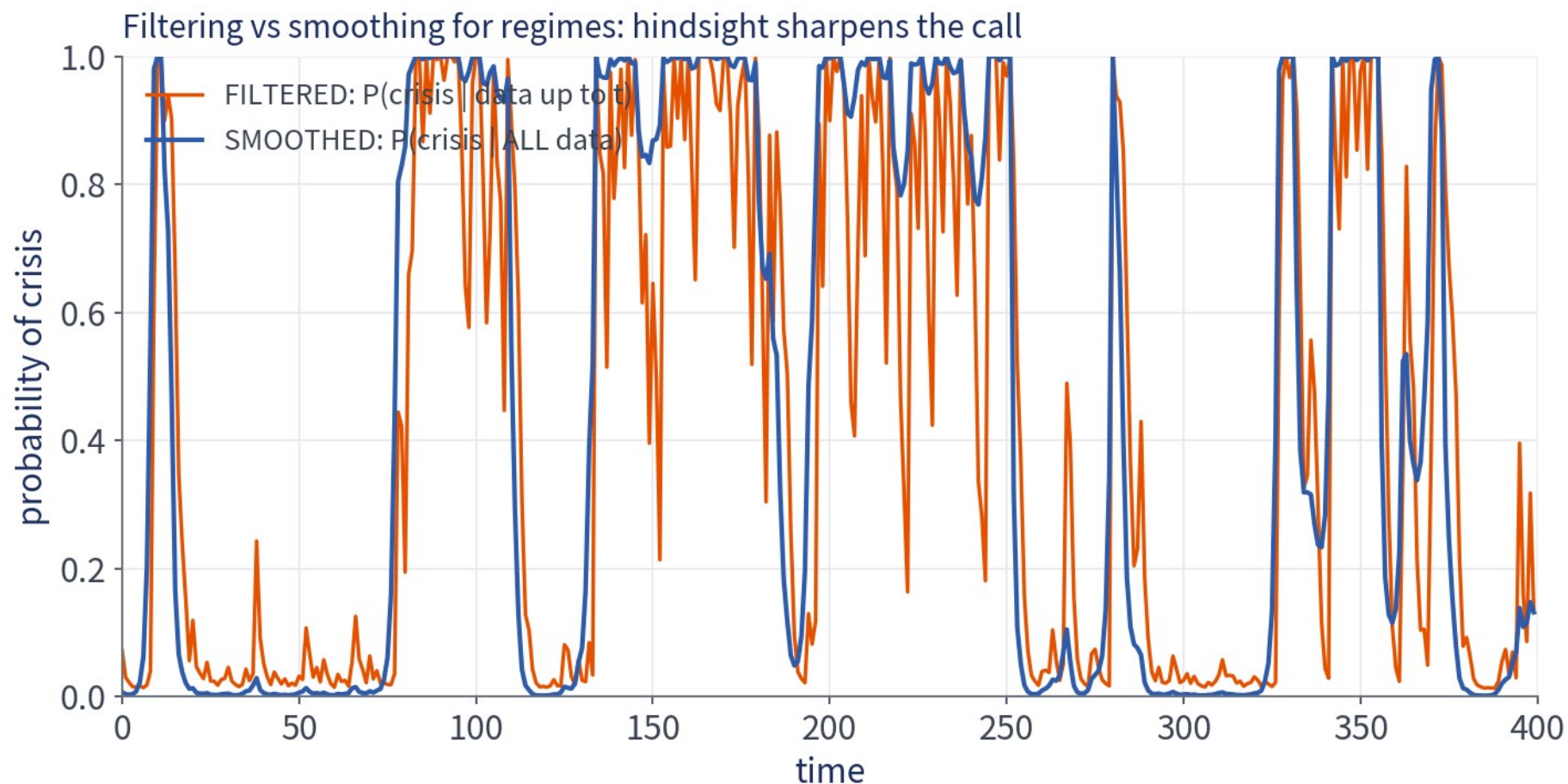
- The regime probability using data up to now.
- Causal and real-time — usable live.
- From the forward pass alone.

Smoothing

- The regime probability using the WHOLE sample.
- A backward pass is added to the forward one.
- Cleaner and more confident, but only in hindsight.

Filtered vs Smoothed Regimes

The filtered probability (orange) reacts in real time but wavers; the smoothed probability (blue), using all data in hindsight, is sharper and more decisive about when each regime held.



Learning: Fitting the Model

Estimating the parameters from data

So far we assumed the HMM's parameters were known. Learning is the problem of estimating them — the transition matrix and each regime's emission — from data in which the states themselves are never observed.

What we estimate

- The initial distribution over regimes.
- The transition matrix between regimes.
- Each regime's emission (its mean and volatility).

The difficulty

- The states are hidden, so we can't just count transitions.
- We must infer the states and the parameters together.
- This calls for an iterative approach.

The Chicken-and-Egg Problem

States need parameters; parameters need states

The circularity

- To estimate parameters, you'd want to know the regimes.
- To infer the regimes, you need the parameters.
- Neither is available at the start.

The resolution

- Guess parameters; infer regime probabilities; re-estimate; repeat.
- Each round improves the fit.
- This alternation is the EM algorithm.

Baum-Welch (EM)

Fitting when the states are hidden

Baum-Welch (EM): E-step: forward-backward $\rightarrow$ M-step: re-estimate (π, A, B)

The two steps

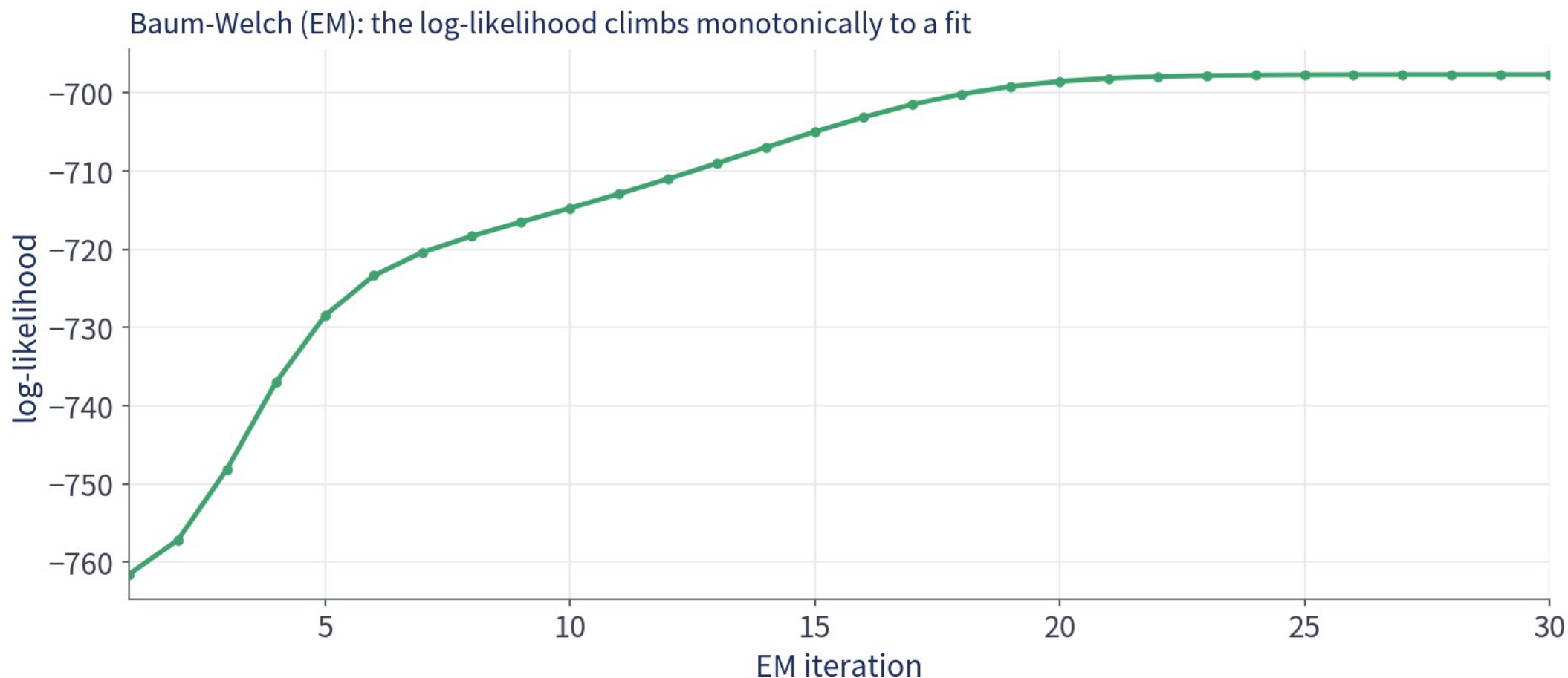
- E-step: run forward-backward to get regime probabilities.
- M-step: re-estimate the parameters using those probabilities.
- Alternate until the fit stops improving.

The guarantee

- Each iteration never decreases the likelihood.
- It climbs to a (local) maximum.
- It is the standard way to train an HMM.

EM Climbs the Likelihood

From a poor initial guess, each Baum-Welch iteration raises the log-likelihood, converging to a fitted model. The monotone climb is the algorithm's defining guarantee.



Forward, Viterbi, and Baum-Welch

Direct Claude Code through all three algorithms

GOAL Run the three HMM algorithms on a return series and see the likelihood, the decoded path, and the fit.

[INPUT] Prompt · Claude Code

Write and run `hmm_algorithms.py` that, on a 2-state Gaussian HMM fit to daily returns:

1. runs the FORWARD algorithm and reports the log-likelihood and the filtered $P(\text{crisis})$;
2. runs VITERBI and plots the decoded regime path;
3. compares FILTERED vs SMOOTHED regime probs;
4. runs BAUM-WELCH from a poor start and plots the log-likelihood climbing each iteration.

Confirm the likelihood never decreases across EM iterations, and that Viterbi gives coherent spans.

How to work with it

- Forward gives probabilities; Viterbi gives a path.
- Smoothed should be sharper than filtered.
- EM's log-likelihood must be monotone up.
- Watch for label switching between runs.

Deliverable: `hmm_algorithms.py` + likelihood, a decoded path, and an EM convergence plot.

Three Questions Before You Move On

If you can answer these, Part 2 has landed

Q1

What does the forward algorithm compute, and why not enumerate paths?

**Likelihood; paths
explode**

Q2

Viterbi differs from the forward pass by doing what?

Max instead of sum

Q3

What is guaranteed to happen to the likelihood in Baum-Welch?

It never decreases

All three come from the forward recursion, Viterbi's max, and EM's monotone climb.

Three Misconceptions to Avoid

Where HMM algorithms trip people up

Common myths

- "Stringing together the per-day best regimes gives Viterbi."
- "Baum-Welch finds the global best parameters."
- "Filtering and smoothing give the same regime call."

The reality

- That pointwise path can be incoherent; Viterbi optimizes jointly.
- EM only reaches a LOCAL maximum — initialization matters.
- Smoothing uses future data, so it is sharper than filtering.

Three Things to Carry Forward

Part 2 in one screen

1

Evaluation — the forward algorithm

It computes the data's likelihood without enumerating exponentially many paths, and its normalized form gives the filtered probability of each regime in real time.

2

Decoding — Viterbi

It finds the single most likely regime path by taking a max instead of a sum and backtracking. That coherent path is how we read regimes off the returns.

3

Learning — Baum-Welch (EM)

Alternating a forward-backward E-step with a re-estimation M-step fits the parameters when the states are hidden, climbing the likelihood to a local maximum.

Next — Part 3: how many regimes, regime-aware pairs trading (Stage IV), and the cautions.

★ WEEK 10 · PART 2 · PRACTICE

Forward and Viterbi, Worked by Hand

Detect a Regime Switch in Real Time; Decode the Best Path

The two core HMM algorithms, run over a return sequence

Part 2's algorithms, with the arithmetic shown. We run the forward recursion over three returns to detect a regime switch as it happens, then run Viterbi on the same data to decode the single most likely regime path.

2P

PART 2 · APPLICATIONS

Sum for the Belief, Max for the Path

Case A: the forward algorithm, chaining regime beliefs over a return sequence to detect a switch in real time. Case B: Viterbi, decoding the single most likely regime path from the same data. Then a Claude Code lab.

What Part 2 Told Us

Now we run two of them - with numbers

1

Evaluation, by the forward algorithm

How likely is this data, and what regime are we in now? Chain the beliefs forward, summing over paths.

2

Decoding, by Viterbi

Which single regime sequence best explains the data? The same recursion, but keep the best path, not the sum.

3

Learning, by Baum-Welch

Fit the transition and emission parameters when the states are hidden. We flag it here and use its output.

Everything here is a calculator exercise: one HMM, three returns, two recursions.

How These Cases Work

Data, then arithmetic, then meaning

The shared setup

- The calm/turbulent HMM from Part 1.
- Three returns: +0.8, then -2.5, then -5.0.
- One calm day, then two bad ones.

What you should watch

- How the forward belief climbs toward turbulent.
- How Viterbi commits to a single path.
- How sum and max give different answers.

The Forward Algorithm

Detect the switch as it happens

Part 1 did one Bayes step; the forward algorithm chains it over a sequence. At each return, we carry every regime's belief forward through the transitions, weight it by how well that regime explains the return, and read off the filtered probability we're in each regime - detecting a switch in real time

The recursion

- Carry each regime's forward score ahead.
- Push it through the transition matrix.
- Weight by the emission likelihood.

What we'll do by hand

- Initialize from the prior and first return.
- Step through returns two and three.
- Read the filtered turbulent probability.

The Forward Recursion

Carry every path forward, summed

$$\alpha_t(j) = \left[\sum_i \alpha_{t-1}(i) a_{ij} \right] b_j(o_t) \quad (\text{carry every path forward, summed})$$

The two moves

- Sum over where you could have come from.
- That's the transition part, in brackets.
- Then multiply by how likely this return is.

Why it's efficient

- It never enumerates full paths.
- Each step reuses the last step's scores.
- Linear in the length of the data.

Initialize, Then Step Forward

The first two forward scores

$$\begin{aligned} \text{init: } \alpha_1(C) &= 0.85(0.38) = 0.324, & \alpha_1(T) &= 0.15(0.11) = 0.017 \\ \alpha_2(T) &= [0.324(0.05) + 0.017(0.80)] \times b_T(-2.5) = 0.0296 \times 0.117 = 0.00347 \end{aligned}$$

Initialize from the prior times the first emission. Then the turbulent score at step 2 sums both ways in, weighted by how well turbulence explains the -2.5% return.

The Forward Table

Scores, and the filtered probability

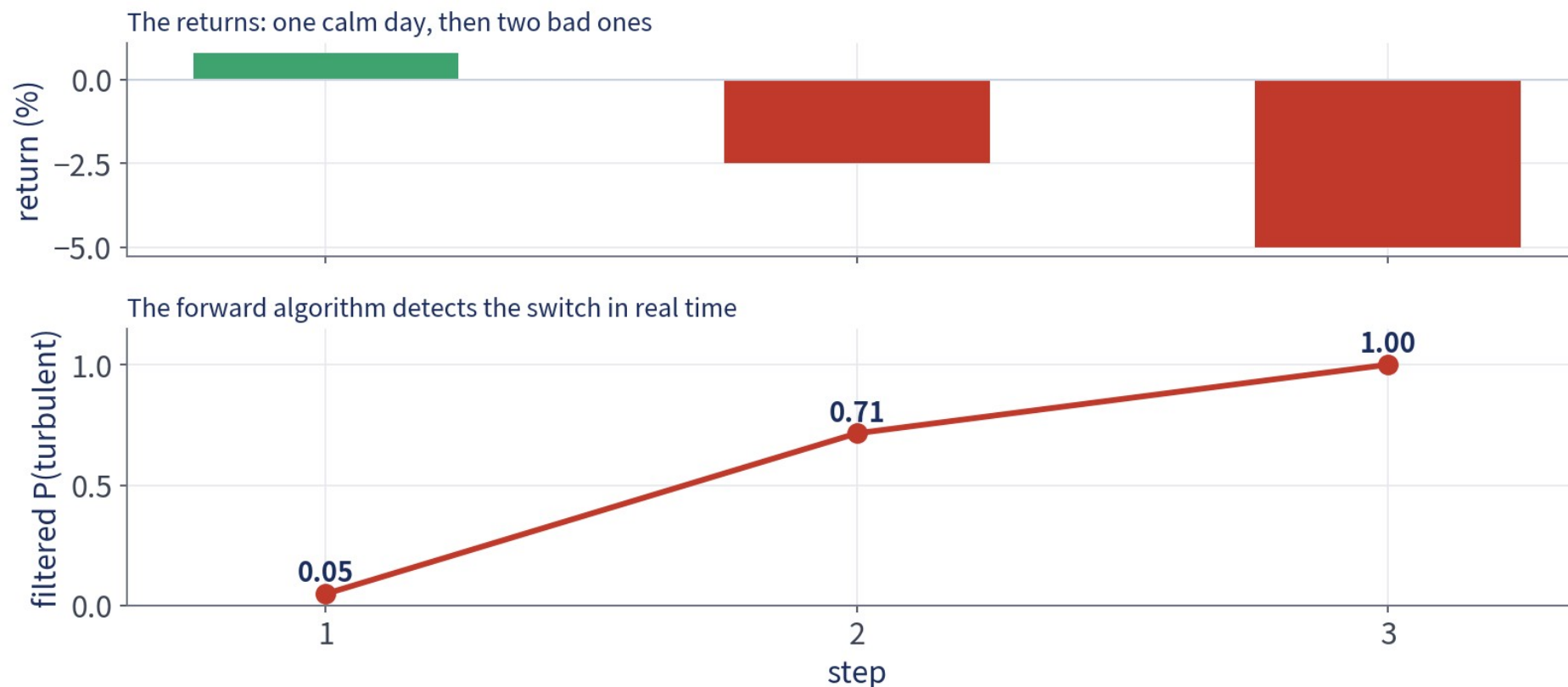
step	return	alpha (Calm)	alpha (Turb)	P(Turbulent data)
1	+0.8	0.324	0.017	0.05
2	-2.5	0.00138	0.00347	0.72
3	-5.0	~0	0.000155	1.00

Normalizing each row gives the filtered probability. It climbs 0.05 -> 0.72 -> 1.00 - the forward pass detects the regime switch as the bad returns arrive.

The Belief Climbs in Real Time

One calm day keeps the turbulent probability near 5%. The first -2.5% return pushes it past 70%; the -5.0% return pins it at essentially 100%. The forward algorithm turns a stream of returns into a live, updating read on the regime

Case A: filtered regime probability climbs 0.05 -> 0.72 -> 1.00



What the Forward Pass Gives

A live regime probability, and a likelihood

The filtered probability

- A real-time read on the current regime.
- It updates the instant a return arrives.
- This is regime filtering, discrete-state.

The likelihood byproduct

- Summing the last column gives $P(\text{data} \mid \text{model})$.
- That's how you compare or fit HMMs.
- One pass, two useful outputs.

The Viterbi Algorithm

Decode the single best regime path

The forward algorithm gives a probability at each step; Viterbi answers a different question: what one regime sequence best explains the whole series? It runs the same recursion but keeps the single best path into each state rather than summing - then backtracks to recover the labels

The difference

- Forward sums over all paths.
- Viterbi keeps only the best one.
- Max instead of sum, plus back-pointers.

What we'll do by hand

- Run the delta recursion over the returns.
- Track which prior state was best.
- Backtrack to the most likely path.

Sum for Belief, Max for the Path

One data set, two questions

forward: $\sum_i$ (all paths, soft probability) Viterbi: $\max_i$ (one path, hard labels)

Forward (sum)

- Soft probabilities at each step.
- Answers 'how likely is each regime now?'
- Marginals, one step at a time.

Viterbi (max)

- One hard sequence of labels.
- Answers 'what path best explains it all?'
- A single global decode.

The Viterbi Recursion

Keep the single best path

$$\delta_t(j) = \max_i [\delta_{t-1}(i) a_{ij}] b_j(o_t) \quad (\text{keep the single best path, maxed})$$

The recursion

- Take the max over where you came from,
- not the sum - the best predecessor only.
- Then weight by the emission, as before.

The back-pointers

- At each step, record which prior state won.
- They're the trail of breadcrumbs.
- Backtracking them recovers the path.

The Viterbi Table

Best scores and back-pointers

step	delta (Calm)	delta (Turb)	best prev -> Calm	best prev -> Turb
1	0.324	0.017	-	-
2	0.00136	0.00190	Calm	Calm
3	~0	0.0000832	Calm	Turb

At each step keep the higher-scoring way in and remember it. At step 3, the best score is turbulent, and its back-pointer leads back through turbulent then calm.

Backtrack to the Path

Follow the back-pointers home

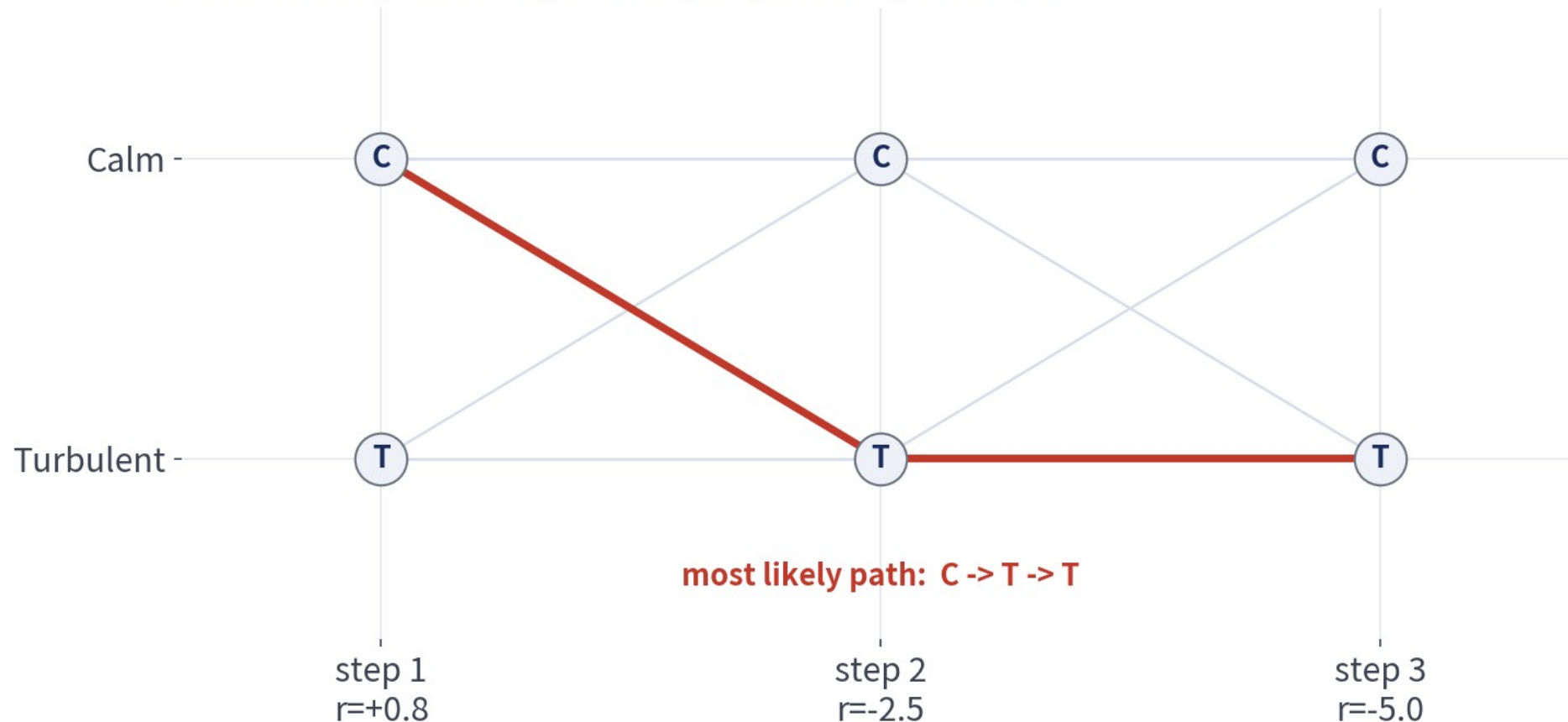
$$\text{best last state} = T \xrightarrow{\text{back-pointer}} t_2 : T \xrightarrow{\text{back-pointer}} t_1 : C \Rightarrow \text{path} = \mathbf{C}, \mathbf{T}, \mathbf{T}$$

Starting from the best final state and following the back-pointers gives Calm, Turbulent, Turbulent - the single most likely regime history for these three days.

The Best Path Through the Trellis

Every step has two possible regimes and every pairing is a possible transition. Viterbi finds the single red path - Calm, Turbulent, Turbulent - that maximizes the probability of the whole sequence, committing to one clean regime history.

Case B: Viterbi picks the single best regime path through the trellis



Why Decode a Hard Path

Labels for backtesting and attribution

When you want the path

- To label a regime history for a backtest.
- To attribute P&L to calm vs turbulent days.
- When a single clean story is what you need.

The subtlety

- The best path isn't the most likely state each step.
- Stringing per-step winners can be inconsistent.
- Viterbi keeps the whole sequence coherent.

Two Recursions, One Trellis

Sum, or max

Both algorithms sweep the same trellis with the same shape of recursion. Swap the sum for a max and you go from a soft, filtered probability to a single hard path. Learning - Baum-Welch - sits on top, using these passes to re-estimate the transition and emission parameters from data

Forward - sum

- Soft regime probabilities in real time.
- The data likelihood as a byproduct.
- Filtering, for discrete states.

Viterbi - max

- One most-likely regime path.
- Clean labels for the whole series.
- Decoding, in a single sweep.

Reproduce Forward and Viterbi

Direct Claude Code to check your arithmetic

GOAL Verify both recursions on the three returns, then run them on a real series.

[INPUT] Prompt · Claude Code

```
Write and run part2_practice.py that, for the
calm/turbulent HMM and returns [0.8, -2.5, -5.0]:
FORWARD:
  * compute alpha at each step; print the filtered
    P(turbulent) and the total data likelihood;
VITERBI:
  * compute delta and back-pointers each step;
  * backtrack and print the most likely path.
Then fit an HMM (Baum-Welch) on a real return
series and decode its regime path.
Match every printed number to the slides.
```

Check against the slides

- filtered $P(T)$: 0.05, 0.72, 1.00.
- likelihood ~ 0.000155 .
- Viterbi path: C, T, T.
- sum $\rightarrow$ soft, max $\rightarrow$ hard.

Deliverable: part2_practice.py whose printed numbers match the forward table and the Viterbi path.

Three Questions on the Cases

If you can answer these, the practice has landed

Q1 What does the forward algorithm compute?

**Filtered regime
probability + data
likelihood**

Q2 How does Viterbi differ from forward?

**It maxes over paths
instead of summing**

Q3 Why not just take the best state each step?

**The path can be
incoherent - Viterbi
keeps it whole**

All three come from the forward table, the Viterbi decode, and the sum-versus-max contrast.

Three Pitfalls to Avoid

Where the algorithms trip people

Common mistakes

- "Enumerate every path to get the likelihood."
- "Viterbi and per-step filtering must agree."
- "Forget to normalize the forward scores."

The reality

- The forward recursion avoids enumeration entirely.
- The best path can differ from per-step winners.
- Normalize to read a probability - or rescale to avoid underflow.

Two Algorithms to Remember

Part 2 applications in one screen

1

Forward detects the switch in real time

Chaining the belief over returns $[+0.8, -2.5, -5.0]$, the filtered turbulent probability climbed $0.05 \rightarrow 0.72 \rightarrow 1.00$, and summing the last scores gave the data likelihood. One pass, a live regime read plus a likelihood.

2

Viterbi decodes the single best path

The same recursion with max instead of sum, plus back-pointers, decoded the most likely regime history: Calm, Turbulent, Turbulent - a clean sequence for labeling and attribution.

3

Sum for belief, max for the path

The forward pass gives soft, filtered probabilities; Viterbi gives one hard, coherent path. Baum-Welch sits on top, using these sweeps to learn the parameters from data.

Next — Part 3 practice: regimes in action, conviction sizing, and Stage IV of the pairs strategy.

★ WEEK 10 · PART 3 of 3

Regimes in Practice

How Many States, and Trading With Them

Choosing the number of regimes · Stage IV · the cautions

Fitting an HMM raises real choices: how many regimes, how to read them live, and how to act on them. We answer these, add regime-awareness to the pairs trade, and confront the traps of latent states.

3

PART 3 · IN PRACTICE

Choosing and Using Regimes

How many states to fit, reading regimes live for trading, pairs trading Stage IV that stands down in crises, and the cautions that come with latent-state models.

From Algorithms to Decisions

Parts 1-2 built it; Part 3 uses it well

1

We can fit and decode HMMs

Forward for likelihood, Viterbi for the path, Baum-Welch to learn. The machinery is in place.

2

Now come the practical choices

How many regimes? How do we read them in real time without look-ahead? And how should a strategy act on them?

3

And the pairs thread gains a fourth stage

Stage IV: use the regime to decide WHEN to trade — standing down when the equilibrium is likely to break.

Part 3 in one line: choose the regimes wisely, trade them carefully, and respect the caveats.

How Many Regimes?

A modeling choice, not a fact of nature

The number of hidden states is not given by the data — you choose it. Too few and the model is crude; too many and it overfits, inventing regimes that are just noise. The count is a judgment call.

The trade-off

- More states fit the training data better, always.
- But extra states can capture noise, not structure.
- The goal is the fewest states that tell a real story.

How to decide

- Penalized likelihood (AIC/BIC) balances fit and complexity.
- Interpretability: can you name each regime?
- Out-of-sample stability of the fitted regimes.

Choosing the Number of States

Let a penalized likelihood guide you

$$\text{BIC} = -2 \log L + k \log n \quad (\text{penalize extra states; pick the lowest})$$

How BIC works

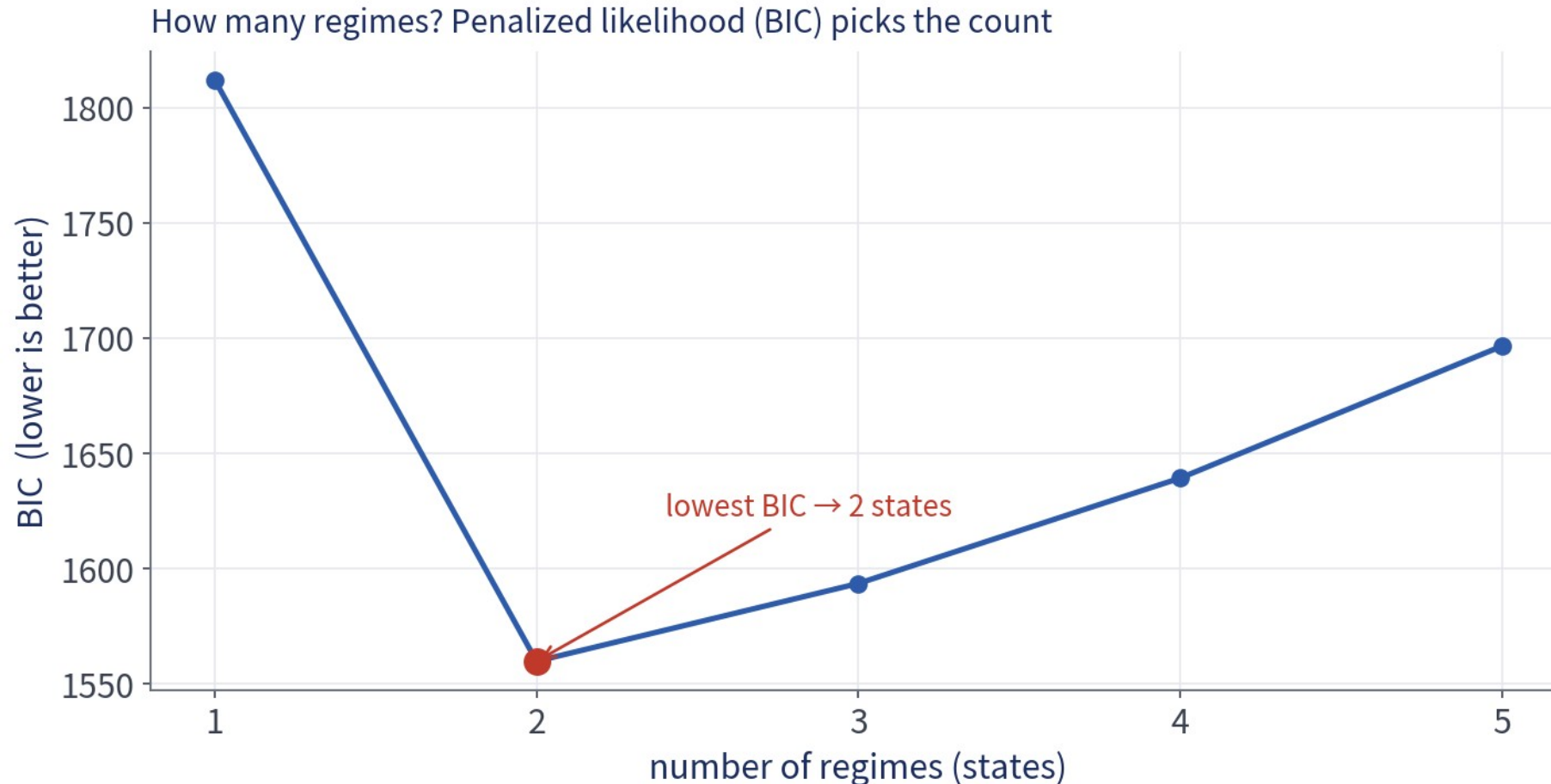
- It rewards fit (higher likelihood) ...
- ... but penalizes each added parameter.
- Pick the number of states with the lowest BIC.

Beyond the number

- Don't follow BIC blindly — check interpretability too.
- A statistically 'better' model with unnamed regimes is suspect.
- Simplicity and stability often win in practice.

How Many Regimes? Ask BIC

Fitting HMMs with one to five states and scoring each by BIC: the criterion drops then rises, and its lowest point suggests the right number of regimes for this data.



The Two-State Default

Calm and crisis, and why it's popular

Why two states

- Calm and crisis is the simplest useful split.
- It is easy to name, interpret, and act on.
- It captures the biggest feature: volatility regimes.

Its strengths

- Robust and hard to overfit.
- The regimes are almost always interpretable.
- A strong default before reaching for more.

When to Use More States

Three or more, and the risks

Sensible extensions

- Three states: bull, bear, and crisis, say.
- Or expansion, slowdown, and recession for macro.
- Use more only when each regime is meaningful.

The risk

- Extra states can splinter into noise regimes.
- Harder to interpret and less stable out of sample.
- Prefer the smallest count that stays interpretable.

Interpreting the Fitted Regimes

Label by behavior, not by index

How to label

- The states come out unlabeled, as 'state 0' and 'state 1'.
- Name them by their fitted mean and volatility.
- High-volatility, negative-mean → the crisis regime.

A caution

- The labels can swap between fits (label switching).
- Always identify regimes by their parameters, not their index.
- Check the transition matrix: which regime is sticky?

Regime Detection in Practice

Turning a fitted HMM into a live read

In production, you don't just decode history — you track the regime as data arrives. The filtered probability of each regime, updated every day, becomes a live gauge of the market's current mood.

The live signal

- Compute the filtered probability of each regime each day.
- Raise an alert when the crisis probability crosses a threshold.
- It updates causally, with no look-ahead.

Why filtered, not smoothed

- Smoothing uses future data — unavailable in real time.
- Live decisions must rely on the filtered estimate.
- Reserve smoothing for after-the-fact analysis.

Filtered vs Smoothed, for Trading

A subtle but crucial distinction

The trap

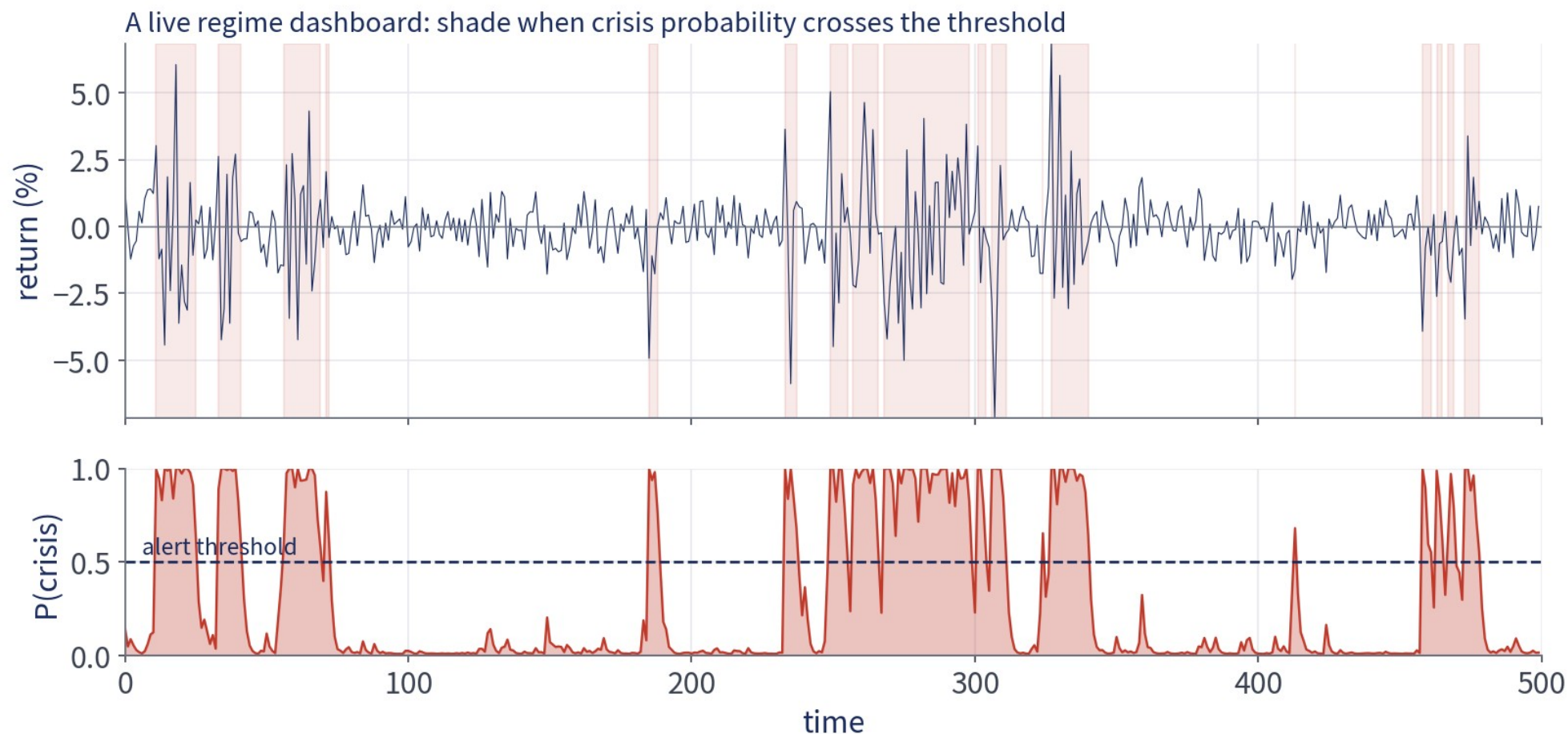
- Smoothed regimes look cleaner and more decisive.
- But they use the whole sample, including the future.
- Back-testing on smoothed regimes is look-ahead bias.

The rule

- Trade on FILTERED regime probabilities only.
- They are what you'd truly have known at the time.
- Use smoothing to explain, never to trade.

A Live Regime Dashboard

The filtered probability of crisis, updated each day. When it crosses the alert threshold, the periods are flagged (shaded) — a real-time read that never peeks at the future.



Stage IV: Regime-Aware Trading

The thread learns when NOT to trade

The pairs trade has bands (I), an anchor (II), and an adaptive hedge (III). Stage IV adds judgment: use the regime to decide whether to trade at all. In a crisis regime, the equilibrium may break — so stand down.

The addition

- Run a regime HMM alongside the spread.
- In the calm regime, trade the spread as usual.
- In the crisis regime, cut size or pause entirely.

Why it helps

- Cointegration tends to break exactly in crises (Week 7).
- Trading blindly then is how pairs strategies blow up.
- Knowing when to sit out protects the whole strategy.

Trade Only in Favorable Regimes

The regime as a position switch

Stage IV rule: $\text{position} = \text{base} \times P(\text{calm} \mid \text{data}) \Rightarrow \text{stand down in crisis}$

The simplest version

- Scale position size by the probability of calm.
- Full size in clear calm; nothing in clear crisis.
- A smooth dial rather than a hard on/off switch.

Variations

- A hard rule: trade only if crisis probability is low.
- Widen stops or shrink size as risk rises.
- The regime filter becomes a risk overlay.

Why Cointegration Breaks in Crises

The danger Stage IV guards against

What happens in a crisis

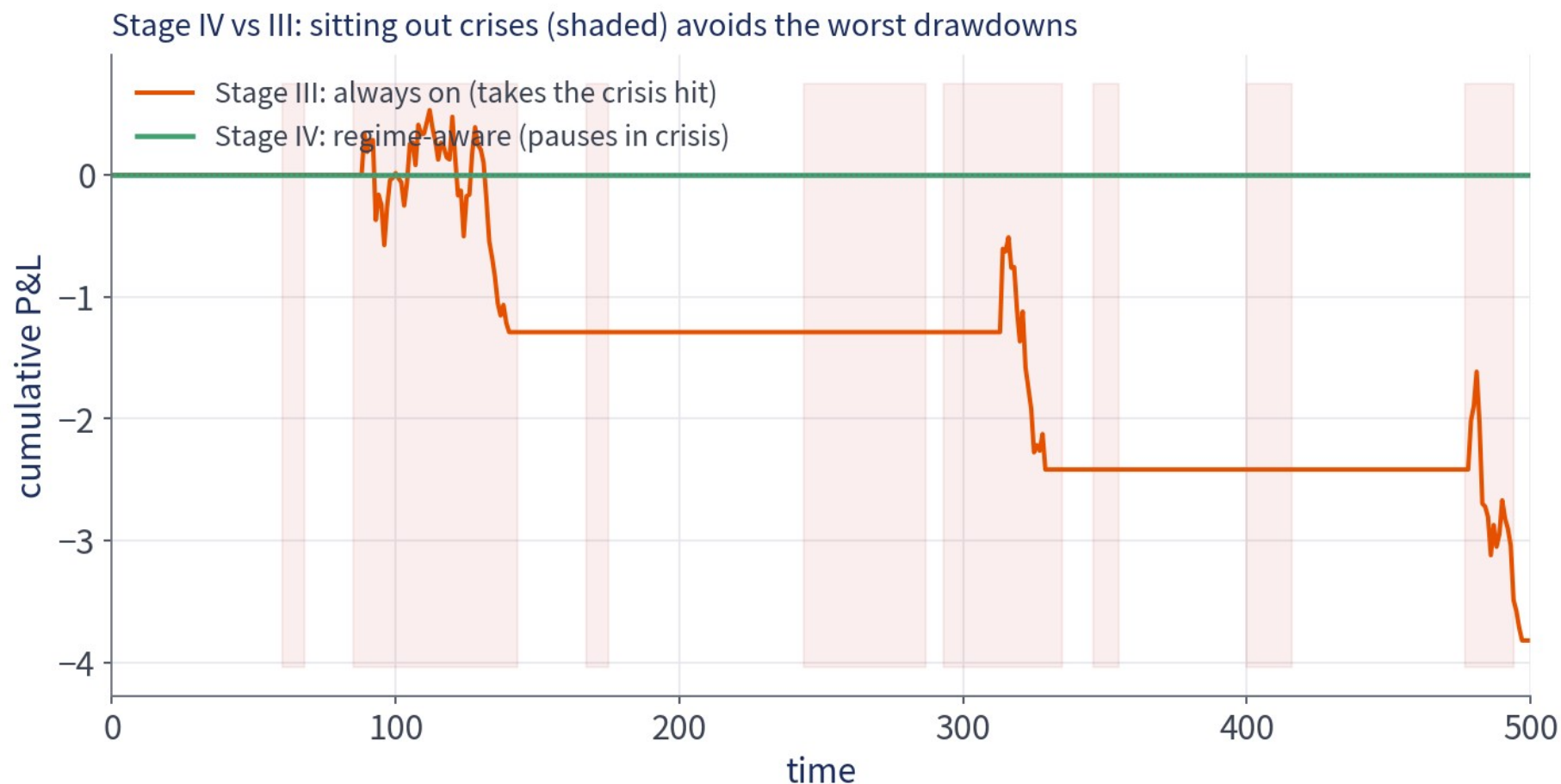
- Correlations and relationships shift abruptly.
- The historical equilibrium can simply stop holding.
- The spread drifts instead of reverting (Week 7).

How the regime filter helps

- It flags the crisis regime as it emerges.
- The strategy stands down before the breakdown compounds.
- It turns a known failure mode into a managed one.

Stage IV vs Stage III

An always-on strategy (orange) takes the full hit when the equilibrium breaks in a crisis (shaded). A regime-aware strategy (green) sits out those periods, avoiding the worst drawdowns.

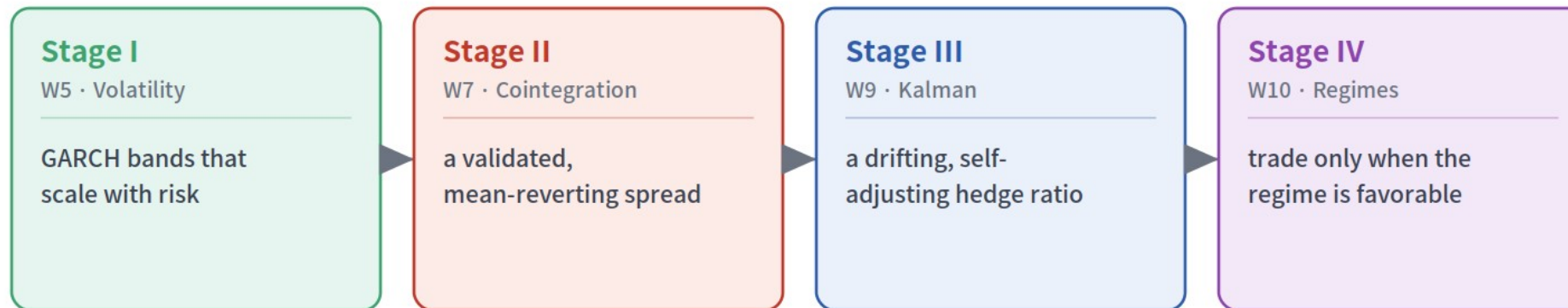


The Pairs Journey, Now Four Stages

Volatility bands (W5), a cointegration anchor (W7), a Kalman-adaptive hedge (W9), and now a regime filter (W10) that decides when to stand down. Knowing when not to trade completes the craft.

The Pairs Journey, Now Four Stages

Each week added a layer; Stage IV adds the wisdom to know when NOT to trade.



The lesson of Stage IV

Cointegration relationships tend to break precisely in crises (Week 7). A regime filter detects that danger and stands the trade down - or sizes it by the probability of calm - instead of trading blindly into a breakdown.

Knowing when NOT to trade is as valuable as the signal itself.

Cautions With HMMs

Latent states are seductive but slippery

An HMM will always find regimes — even in random data. The story it tells is compelling, which is exactly the danger. Latent-state models demand extra skepticism before you trust or trade them.

The seduction

- A fitted HMM produces a clean regime narrative.
- It looks insightful even when it is fitting noise.
- Compelling stories are the easiest to over-trust.

The discipline

- Validate regimes out of sample, not just in-sample.
- Check they are stable and economically sensible.
- Be ready to conclude the regimes aren't real.

Overfitting Regimes

More states, better story, worse model

The symptom

- Adding states keeps improving the in-sample fit.
- New regimes appear that have no real meaning.
- The model memorizes noise as 'structure'.

The guard

- Penalize complexity (BIC) and prize interpretability.
- Insist that regimes persist out of sample.
- When in doubt, use fewer states.

Label Switching and Identifiability

The regimes have no fixed names

The issue

- Which state is 'regime 0' is arbitrary and can flip between fits.
- So results keyed to the index are unstable.
- Two runs may 'disagree' only by relabeling.

The fix

- Identify regimes by their parameters (mean, volatility).
- Sort or pin states by, say, volatility before comparing.
- Never rely on the raw state index.

Regimes Are a Model, Not Reality

The map is not the territory

The reminder

- 'Calm' and 'crisis' are useful fictions, not physical states.
- The market doesn't truly toggle a two-position switch.
- The HMM is a convenient approximation of messier dynamics.

Why it matters

- Don't over-interpret a decoded regime as ground truth.
- Treat regime signals as one input, not gospel.
- Keep the model's limits in view when acting on it.

When an HMM Is the Wrong Tool

Match the model to the dynamics

Poor fits

- A truly continuous, smoothly-drifting state (use Kalman).
- Dynamics with long memory beyond one step.
- Cases where regimes can't be named or validated.

When it shines

- Discrete, persistent regimes like calm vs crisis.
- Volatility clustering with abrupt switches.
- A risk overlay that says when to stand down.

Regime-Aware Pairs Trade (Stage IV)

Direct Claude Code to trade only in calm regimes

GOAL Overlay a regime HMM on a pairs trade and back-test trading only when the calm regime is likely.

[INPUT] Prompt · Claude Code

Write and run `pairs_stage4.py` that:

1. fits a 2-state HMM to the spread's returns (or to a market index) to get FILTERED $P(\text{crisis})$;
2. back-tests the Stage III z-score pairs trade TWICE: always-on, and regime-aware (trade only when $P(\text{crisis})$ is below a threshold);
3. plots cumulative P&L for both, shading crises;
4. reports drawdown and Sharpe for each.

Use FILTERED (not smoothed) regimes to avoid look-ahead. Show the regime overlay cuts drawdown.

How to work with it

- Use filtered regimes only — no future data.
- Regime-aware should have smaller drawdowns.
- Compare Sharpe, not just total P&L.
- This completes the pairs thread (Stages I-IV).

Deliverable: `pairs_stage4.py` + always-on vs regime-aware P&L, with drawdown and Sharpe.

Three Questions Before You Finish

If you can answer these, Week 10 has landed

Q1	How do you choose the number of regimes?	BIC + interpretability
Q2	For live trading, filtered or smoothed regimes — and why?	Filtered (no look-ahead)
Q3	What does Stage IV add to the pairs trade?	Stand down in crises

All three come from model selection, the look-ahead rule, and regime-aware trading.

Three Pitfalls to Avoid

Where regime trading goes wrong

Common mistakes

- "More regimes means a better model."
- "Back-test on the smoothed regime path."
- "The decoded regime is the market's true state."

The reality

- Extra regimes often fit noise; prefer the fewest that work.
- Smoothed uses the future — trade on filtered only.
- Regimes are a useful model, not ground truth.

Week 10 in Three Ideas

Pulling the whole week together

1

A hidden regime, seen through returns

Parts 1-2: a Markov chain of latent regimes emits returns; the forward algorithm, Viterbi, and Baum-Welch evaluate, decode, and learn it — the discrete cousin of the Kalman filter.

2

Choose few regimes, read them live

Part 3: pick the number of states by penalized likelihood and interpretability, and track them with FILTERED probabilities to avoid look-ahead in trading.

3

Stage IV — know when not to trade

A regime filter stands the pairs trade down when the equilibrium is likely to break. But regimes are a model, not reality — trust them with discipline.

Next — Week 11: continuous-time models and stochastic differential equations.

★ WEEK 10 · PART 3 · PRACTICE

Regimes in Action, Worked by Hand

Choose the Number of States; Gate the Trade by Regime

Model selection by BIC, and Stage IV of the pairs strategy

Part 3's practice, with the arithmetic shown. We pick the number of regimes with a penalized likelihood, then put Stage IV to work - using the filtered regime probability to gate the pairs trade and step aside in a crisis.

3P

PART 3 · APPLICATIONS

Choose the Model, Then Gate the Trade

Case A: compare one-, two-, and three-state HMMs by BIC and let the penalty pick two. Case B: gate the pairs trade on the crisis probability and compare the P&L with and without the gate. Then a Claude Code lab.

What Part 3 Told Us

Now we make it concrete - with numbers

1

How many regimes is a choice

Not a fact of nature. A penalized likelihood like BIC balances fit against complexity - and usually lands on two.

2

Turn a fitted HMM into a live read

Use the filtered probability, not the smoothed one, to avoid look-ahead. It's the honest, real-time regime estimate.

3

Stage IV: trade only when it's safe

Cointegration breaks in crises, so the regime becomes a gate. We compute the P&L it saves, by hand.

Everything here is a calculator exercise: three model scores, five trading days, one gate.

How These Cases Work

Data, then arithmetic, then meaning

Each case has three steps

- Start from fitted scores or a signal.
- Compute the BIC or the gated P&L by hand.
- Then read the model or trading decision.

What you should watch

- How the penalty stops you overfitting regimes.
- How the gate steps aside in a crisis.
- How Stage IV completes the pairs thread.

How Many Regimes?

Let a penalized likelihood decide

More regimes always fit the data better - but at some point you're just fitting noise. BIC scores each model by its fit minus a penalty for its parameters. We compute it for one-, two-, and three-state HMMs and let the lowest score pick the model.

The trade-off

- More states raise the log-likelihood.
- But each state adds parameters to penalize.
- BIC balances the two.

What we'll do by hand

- Count parameters for each model.
- Compute BIC for one, two, three states.
- Pick the lowest.

The BIC Criterion

Fit, minus a penalty for complexity

$$\text{BIC} = -2 \log L + k \ln n \quad (\text{fit reward, minus an overfitting penalty; lower is better})$$

The two terms

- Minus-two-log-L rewards a better fit.
- $k \ln(n)$ penalizes more parameters.
- Lower BIC is better overall.

Why it works

- A richer model must earn its extra parameters.
- If the fit gain is small, the penalty wins.
- It guards against overfitting regimes.

BIC for One, Two, Three States

Fit gains, penalty grows

$$\text{1-state : } -2(-820) + 2(6.22) = 1652$$

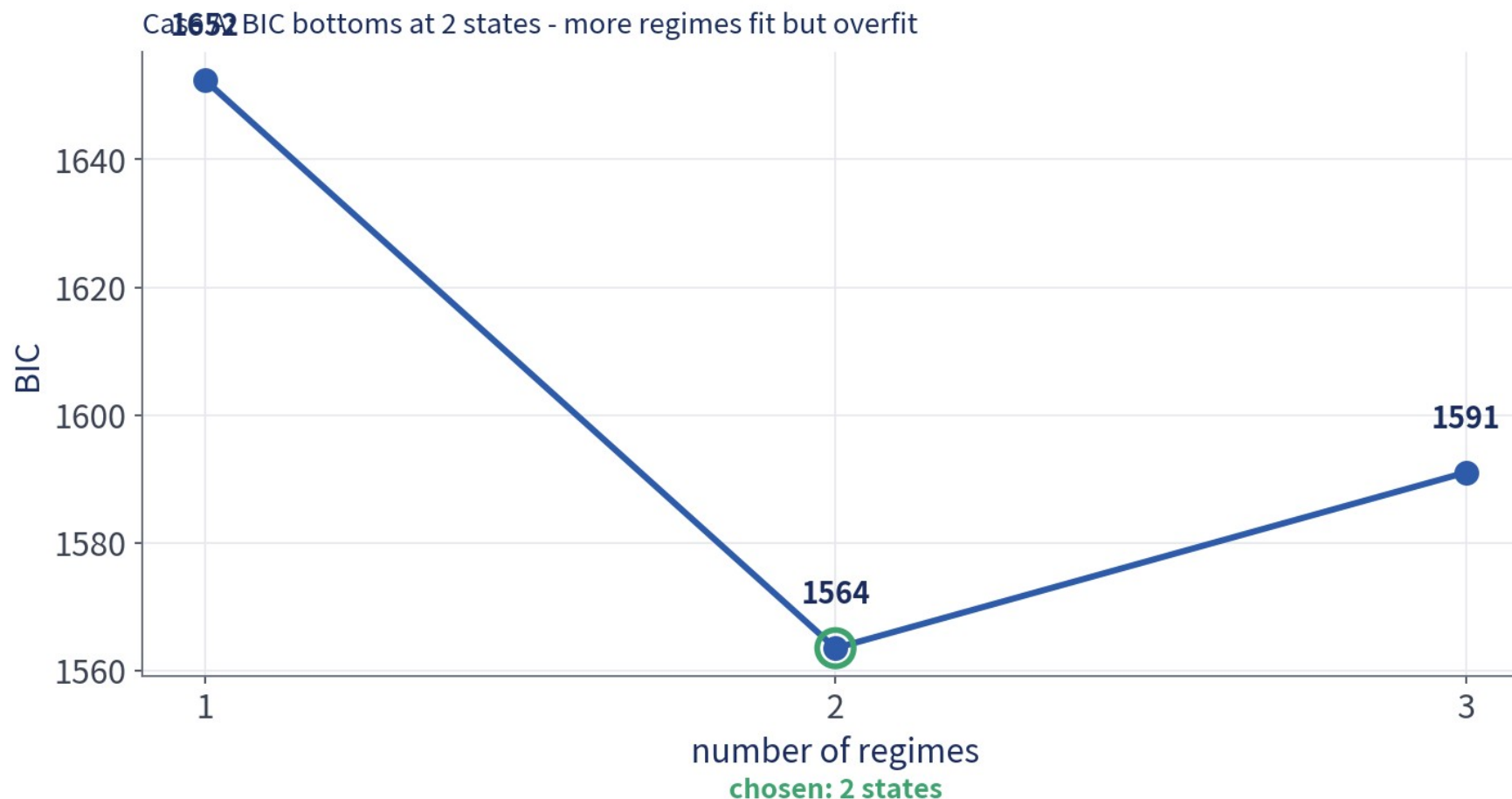
$$\text{2-state : } -2(-760) + 7(6.22) = \mathbf{1564} \quad (\text{lowest})$$

$$\text{3-state : } -2(-752) + 14(6.22) = 1591$$

Three states fit slightly better than two (log-L -752 vs -760), but the extra parameters cost more than they're worth - so the two-state model has the lowest BIC.

BIC Bottoms at Two States

The BIC falls sharply from one to two states as the fit improves, then rises again at three as the penalty for extra parameters overtakes the small gain. The curve's minimum - two states - is the calm/crisis default, chosen not by taste but by the score



Don't Overfit the Regimes

Two states is usually enough

The temptation

- More regimes always look better in-sample.
- You can label every wiggle its own state.
- It won't generalize.

The discipline

- Let BIC penalize the extra parameters.
- Two states - calm and crisis - is the default.
- Interpretable, robust, and usually right.

Stage IV: Gate the Trade

The thread learns when NOT to trade

Cointegration - the anchor of the pairs trade - breaks down in a crisis, exactly when the spread blows out. Stage IV uses the fitted HMM as a gate: trade the spread only when the crisis probability is low. We compute the P&L with and without the gate over five days.

The setup

- Five days of spread-trade P&L.
- The filtered crisis probability each day.
- Two of the days are a crisis.

What we'll do by hand

- Apply the gate: trade only if crisis is unlikely.
- Sum the gated and ungated P&L.
- See what the gate saved.

The Regime Gate

A position switch driven by the regime

$$\text{position}_t = \text{signal}_t \times \mathbf{1}[P(\text{crisis})_t < 0.5] \quad (\text{trade only in a favorable regime})$$

The rule

- Take the signal only in a calm regime.
- When crisis is likely, stand flat.
- The regime is an on/off switch.

Why filtered, not smoothed

- You must decide in real time.
- Smoothed uses future data - look-ahead.
- Use the filtered probability to be honest.

Five Days, Gated

Trade only when crisis is unlikely

day	spread P&L	P(crisis)	gate	gated P&L
1	+0.4	0.10	ON	+0.4
2	+0.3	0.15	ON	+0.3
3	-2.5	0.80	OFF	0.0
4	-1.8	0.70	OFF	0.0
5	+0.5	0.20	ON	+0.5

The gate is OFF on days 3 and 4, when the crisis probability is high - exactly the days the spread trade lost badly. Those losses are avoided entirely.

With and Without the Gate

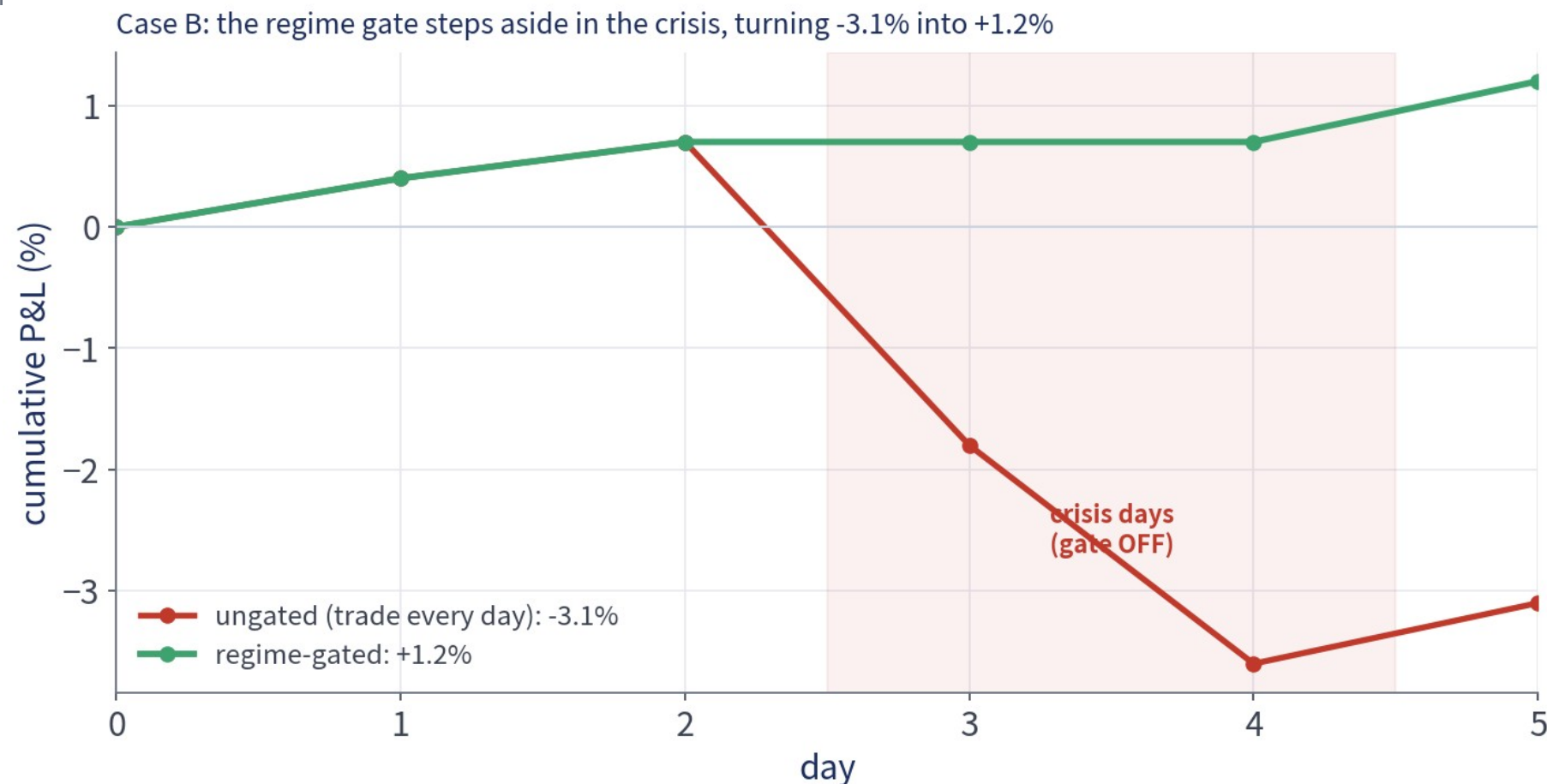
The P&L the regime saved

$$\begin{aligned}\text{ungated} &= 0.4 + 0.3 - 2.5 - 1.8 + 0.5 = -\mathbf{3.1\%} \\ \text{regime-gated} &= 0.4 + 0.3 + 0 + 0 + 0.5 = +\mathbf{1.2\%}\end{aligned}$$

Trading every day loses 3.1%; gating on the regime turns it into a 1.2% gain - a 4.3-point swing, purely from standing aside in the crisis.

The Gate Steps Aside in the Crisis

The ungated curve (red) collapses through the shaded crisis days as the spread blows out. The gated curve (green) simply goes flat there, having read the regime, and finishes positive. Stage IV's value is the loss it refuses to take



Hard Gate or Soft Sizing?

On/off, or scale by conviction

The hard gate

- Fully on below 0.5 crisis, fully off above.
- Simple, and it cleanly dodged the crisis.
- But it's sensitive to the threshold.

Soft conviction sizing

- Scale the position by $P(\text{calm})$ instead.
- Smoother, but keeps some crisis exposure.
- Here it netted about zero - the hard gate won.

Why Stage IV Matters

Knowing when not to trade

The danger it guards

- Cointegration breaks in a crisis.
- The spread blows out, not reverts.
- That's when the trade loses most.

The edge it adds

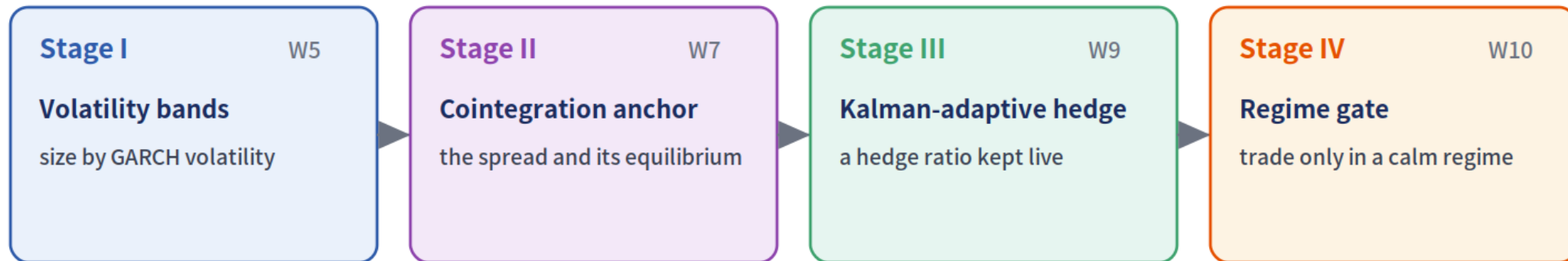
- The HMM sees the regime shift coming.
- It turns the trade off before the damage.
- Not trading is itself a decision.

The Pairs Journey, Now Four Stages

Each week added one piece: volatility bands size the trade (W5), cointegration anchors the spread (W7), a Kalman filter keeps the hedge live (W9), and now a regime gate decides whether to trade at all (W10). Stage IV is the discipline of standing aside.

The Pairs Journey, Now Four Stages

Each week added one piece; Stage IV teaches the strategy when NOT to trade.



Stage IV is the gate

Cointegration breaks in crises - so the HMM turns the trade OFF when the crisis regime is likely.

Reproduce the Selection and the Gate

Direct Claude Code to check your arithmetic

GOAL Verify the BIC choice and the regime-gated P&L, then backtest the gate.

[INPUT] Prompt · Claude Code

Write and run `part3_practice.py` that:

CASE A - model selection:

- * for $\log-L = -820, -760, -752$ and $n = 500$, compute k and BIC for 1, 2, 3 states; print the winner.

CASE B - the gate:

- * from the 5-day P&L and crisis probabilities, compute ungated and regime-gated totals;
- * also try soft sizing by $P(\text{calm})$.

Then fit a 2-state HMM on a real spread and backtest the gated strategy vs always-on.

Match every printed number to the slides.

Check against the slides

- BIC 1652, 1564, 1591.
- chosen: 2 states.
- ungated -3.1%, gated +1.2%.
- soft sizing ~ 0 .

Deliverable: `part3_practice.py` whose printed numbers match the BIC choice and the gated P&L.

Three Questions on the Cases

If you can answer these, the practice has landed

- Q1** Why not just pick the most regimes?
-
- Q2** Why gate the pairs trade on the regime?
-
- Q3** Filtered or smoothed probability for the gate?
-

BIC penalizes parameters - it overfits

Cointegration breaks in a crisis

Filtered - smoothed peeks at the future

All three come from the BIC selection and the Stage IV regime gate.

Three Pitfalls to Avoid

Where regime trading slips

Common mistakes

- "More regimes is a better model."
- "Use the smoothed regime to trade live."
- "A fitted HMM is a crystal ball."

The reality

- BIC penalizes complexity - two states usually wins.
- Smoothed uses the future - only filter is honest live.
- It's a probability, not a guarantee - size accordingly.

Two Calculations to Remember

Part 3 applications in one screen

1

Let BIC choose the number of regimes

Three states fit marginally better (log-L -752 vs -760), but with BIC 1591 versus 1564 the two-state model wins - the penalty for extra parameters outweighs the small fit gain. Calm and crisis, chosen by the score.

2

Gate the trade on the crisis probability

Trading every day lost 3.1% over five days; gating the trade off when crisis probability exceeded 0.5 skipped the two crisis days and turned it into a 1.2% gain - a 4.3-point swing from not trading.

3

Stage IV completes the pairs thread

Volatility bands, a cointegration anchor, a Kalman-adaptive hedge, and now a regime gate. The strategy has finally learned the hardest discipline: when to stand aside.

Next — Week 11 and continuous-time models: stochastic differential equations enter the story.